Coursework #3 – ASSIGNMENT (40%)

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Learning Outcomes Assessed	
CLO3	Demonstrate capacity to perform a self-directed piece of practical work that requires the application of Natural Language Processing and text mining techniques. (P5, PLO6)

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1. Executive Summary

The sheer increase in the e-commerce platform has produced an unmatched amount of the user-generated text in the form of product reviews. There are hundreds of millions of customer reviews in Amazon alone, and it is operationally infeasible to analyze them manually. This project has met this challenge by designing, implementing and critically evaluating an entire Natural Language Processing (NLP) pipeline to classify sentiment based on 3 classes Negative, Neutral, and Positive using a subset of 568,454 reviews of the Amazon Fine Food Reviews dataset.

The project was designed along the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology because they selected it due to its iterative business-driven approach which meant that all technical decisions were kept in line with a very well-defined business task: facilitating organisations easily monitor customer satisfaction, detect product quality problems, and either direct feedbacks at scale without the need to have a human interaction.

It was designed and coded a strictly regulated preprocessing pipeline, including the elimination of HTML tags, URL dwelling, contraction expansion, special character normalisation, downcasing, word-level tokenisation, stopword elimination, intentional negation retention (ex: not, never, could not, cannot) and WordNet lemmatisation. Several crucial but justified design decisions were deemed necessary by the ability of the researcher to retain the presence of negation words, as the elimination of this presence would essentially ruin the semantic polarity of reviews (e.g., the “not good is still good reduced to good). The dataset had gross imbalance of classes hence positive reviews comprised more than 78 of the total raw corpus. In order to avoid model bias and provide a fair evaluation, the dataset was adjusted to an equal division: 20,000 representatives of any sample class (60,000 in total), the oversampling was used in case of minority classes.

The type of feature extraction was TF-IDF using a range of unigram-bigram and 15,000 features. It is not a coincidence: bigrams encode phrases with feelings e.g. by describing a person as highly recommend, that single-word representations cannot encode. Sublinear term frequency scaling was adopted to diminish the domination of high-frequency terms which have little discriminatory value.

Three classical machine learning algorithms were carried out and the comparison of these algorithms was systematized Multinomial Naive Bayes, Logistic Regression (multinomial) and Linear Support Vector Machine(LinearSVC). Individual models were optimised using the hyperparameters every model using GridSearchCV using 3-Fold Stratified Cross-Validation also to ensure that performance estimates that are reported are robust and unbiased. The best-performing model was Logistic Regression with a weighted F1-score of 0.7487 (the difference between SVM and Naive Bayes was 0.0023 and 0.0381 respectively). Another trend that was found in all models was the distinction of the Neutral reviews that had low recalling than the Negative and Positive classes. This observation can be attributed linguistically, three-star reviews are often full of mixed feelings, ambiguous statements, and comparative clauses that may not be easily clear without contextual basis, which is a weakness of bag-of-words feature descriptions.

Most importantly, despite the fact that all three models underperformed relative to the target weighted F1-score of 0.80, the findings set high baselines and provide a clear path to data

improvement, such as the incorporation of contextualised word embeddings (e.g., BERT), the utilisation of larger balanced datasets, and the search of in-depth methods of data integration. The deployment stage showed the practical relevance of the pipeline by the real-time sentiment prediction feature that was able to categorize the unknown reviews providing readable results.

2. Introduction & NLP Problem Statement

2.1 Background

The paradigm shift in digitalization of retailing commerce has radically influenced the interaction of the consumers with products and brands. Alongside traditional transactional online websites like Amazon, this uses online marketplaces as a not only transactional website but a huge storehouse of consumer opinion, where millions of product reviews are being posted every day and at thousands of different categories. This customer generated information is one of the most resource-rich and actionable sources of business intelligence that organisations have at their disposal, but the sheer amount of it makes it next to impossible to interpret it manually.

A sub-discipline of artificial intelligence and computational conception of human language, which is Natural Language Processing (NLP) is an answer to this dilemma. NLP allows the automated identification of meaningful information within unstructured textual data at a scale and speed impossible to human analysts through the application of text mining and machine learning methods. Such a common and commercially significant NLP task as sentiment analysis is vernierly concerned with the task of ascertaining the emotional orientation or opinion held in a given piece of writing, namely whether the text is positive, negative or neutral.

2.2 Motivation

Although the volume of customer feedback received has a commercial meaning, there is a big disparity between the number of reviews received and the ability of an organisation to respond to it. Going by the studies conducted in opinion mining and sentiment analysis, those businesses who are successful in using customer sentiment information depict statistically significant returns in the form of product enhancement, retention and an increase in income. Nevertheless, the language complexity of review text, which includes casual language, sarcasm, ambivalent views, vocabulary specializing to specific domain, and emotionally colored phrases, poses significant computational issues, which from an engineering perspective is why sentiment classification is not a trivial research and engineering problem.

Amazon Fine Food Reviews data would be best suitable to investigate these issues. It represents more than a decade of consumer review of a specialised category of products, which not only engulfes the wide scope of human sentiment display, including both highly positive expressions and scathing criticism, but also the linguistically grey zone of neutral opinion. This variety gives it the perfect standard to measure the suitability of classical NLP pipelines.

2.3 Problem Statement

This project will solve the following NLP problem that is defined in a formal manner:

"Assuming a natural language review of an unstructured and raw Amazon food product, implement a machine-based system which can specifically predict sentiment as Negative (1-2 stars), Neutral (3 stars), or Positive (4-5 stars) without using the numerical Star rating itself at the time of the prediction."

This definition is deliberately restricted to a text-only classification, which is to capture the real world deployment case where star ratings are not available, are invalid, or are falsified. Three-class classification offers more complexity in comparison with the binary sentiment mailing system allocation, which the Neutral category embody a gamut of uncertain, mixed, or weakly opinionated wording that is naturally hard to differentiate between both polar classes.

2.4 Project Objectives

The project was planned to be directed by the following objectives:

1 Business Objective: Develop an NLP system that can be scaled up to support the monitoring and categorising of customer sentiment on product review data and the subsequent utilisation of this information to help the e-commerce businesses make sound decisions in product development, quality assurance and management of customer experience.

2 Technical Objective: Have a complete NLP preprocessing pipeline that works effectively to convert raw noisy review text into clean and normalised well-machine learnable representations.

3 Modelling Objective: Model, optimise, and benchmark at least three classical machine learning models Multinomial naive bayes, Logistic regression and Linear SVM to determine the most useful model to use in classing the Amazon Fine Food Reviews dataset as having one of three classes.

4 Evaluation Objective: Strictly assess all models with task-relevant measures, such as weighted F1-score, precision, recall, and confusion matrix assessment, and critically discuss them in the framework of the overall performance of the class and business applications.

5 Analytical Objective: Determine the shortcomings of the implemented pipeline and recommend elaborated evidence-based efforts to improve it in the future, such as using advanced deep learning and transformers.

2.5 Significance of the Study

The importance of this project does not end with an academic exercise. A precise automated sentiment classifier can help organisations (in terms of business) to: process thousands of reviews in seconds instead of days; detect emerging product quality problems before they develop into reputational damage; target customer responses; and provide aggregated sentiment trends over product line, over time. Researchwise, the study offered by this project is a methodical comparison of classical methods of NLP using a large scale realistic dataset

under regulated experimental conditions - a baseline against which more advanced methods can be gauged.

2.6 Scope and Constraints

This project is scaled to classical machine learning methods with feature representations by TF-IDF. The models that are intentionally omitted in the implementation part are deep learning models (LSTM, BERT) and lexicon-based approaches, which are mentioned as the way forward. The scope of the analysis is limited to the reviews in English and timewise limited to the scope of the Amazon Fine Food Reviews corpus (October 1999 - October 2012).

3. Literature Review

3.1 Overview

This literature review is investigating the underlying and modern works within the field of sentiment analysis, text classification, NLP preprocessing strategies, and feature extraction strategies that directly apply within the context of this project. The review is thematically structured in order to put the methodological decisions made into context and to position this work in the larger context of research in NLP.

3.2 Sentiment Analysis: Origins and Evolution

Sentiment analysis or opinion mining is a research field that was officially canonized in the early 2000s. Pang, Lee and Vaithyanathan (2002) seminal work proved that machine learning methods, in especially Naive Bayes, Logistic Regression and Support Vector, could successfully be used to characterize or categorize the sentiment polarity of a movie review with an accuracy level of up to 82.9 percent using the Support Vector Machines. This research paper was initially the basis of making supervised machine learning opinion classification viable, and became the standard methodology upon which future studies were established.

Subsequently, Liu (2012) offered a bit of a taxonomic scheme of sentiment analysis, differentiating the document level, sentence level, and aspect level sentiment classification. The paradigm used in this project is document-level classification, which assigns one film-level sentiment label to a whole review, and it is still the most researched formulation because there are labelled datasets, like star ratings, which are natural proxy measures of overall sentiment polarity.

Medhat, Hassan, and Korashy (2014) systematically reviewed sentiment analysis algorithms and applications, which identified more than 54 research articles, and found machine learning methods as always better than lexicon-based ones on domain-specific data sets. Their results are mostly applicable in this project when selecting supervised classification as opposed to rule-based models.

3.3 Sentiment Analysis on Product Reviews

The selected field of product review affective analysis has received significant amounts of research both because of the business importance of online shopping websites. Hu and Liu (2004) were the first to have applied aspect-based sentiment analysis to Amazon product reviews, establishing that the opinions of customers could be broken down into feature-relevant sentiments - the prototype of more fine-grained NLP tools in e-commerce.

In a different study, McAuley and Leskovec (2013) also analyzed the same dataset used in the current project (Amazon Fine Food Reviews) studying the correlation between the review quality, the helpfulness votes and the expression of sentiment. Their results showed that the helpfulness of review is heavily related to linguistic sophistication, and food product review has patterns of vocabulary positions that could be said to be local to domains, compared with the food product review compared with general-purpose review corpora.

Agarwal et al. (2011) analyzed the multi-class sentiment classification in Twitter data and observed that it is significantly more difficult to classify data as three classes (positive, negative, neutral) than as binary since the neutral category is linguistically ambiguous. This observation was the basis of the expectation management in this project where the Neutral classes performance was expected to be low compared to the Negative and Positive classes.

3.4 Classical Machine Learning Approaches for Text Classification

3.4.1 Naive Bayes

The Naive Bayes classifiers possess a long history of use in the field of text classification because they are computationally efficient and possess surprisingly good performance despite the severe conditional independence assumption. As demonstrated by Rennie et al. (2003), Multinomial Naive Bayes operating on TF-IDF weighted features was capable of being equal or better than more complex classifiers on standard text benchmarks. The simplicity and interpretability of the classifier renders it common baseline in NLP research a task it achieves in this project.

3.4.2 Logistic Regression

The Logistic Regression has always been found to be among the strongest in terms of high dimensional text classification. Wang and Manning (2012) made an extensive comparison of Naive Bayes and Logistic Regression on several NLP benchmarks, and they found that Naive Bayes was comparatively less competitive with longer texts but Logistic Regression is more competitive with longer texts when using TF-IDF features and bigrams. This observation would be consistent with the outcomes of this project, in which Logistic Regression had the best weighted F1-score of 0.7487 on the Amazon review corpus, which has reviews of different length and tends to be of moderate and substantial size.

3.4.3 Support Vector Machines

Joachims (1998) brought in the Support Vector Machines to text classification and showed them to be vastly more suitable to separate texts based on high dimensionality, sparse features spaces the very nature of TF-IDF matrices. LinearSVC, a linear variant of the kernel has proven to be especially effective when dealing with text classification since the data is often

linearly separable in high-dimensional TF-IDF space. Mullen and Collier (2004) used SVM in sentiment classification and it was better than the other classical method in combination with linguistic features. In the current project, SVM has reached a weighted F 1-score of 0.7464, which is quite competitive in comparison to the Logistic Regression, a fact supported by the evidence of the related literature.

3.5 Feature Extraction Methods

3.5.1 Bag-of-Words and TF-IDF

Written in the setting of information retrieval, Salton and McGill (1983) introduced the Bag-of-Words (BoW) model, which features documents as unordered words frequencies sets, and throws grammatical structure and word order into the trash. Although it is a semantically limited representation, it has been shown to be incredibly useful in sentiment classification with TF-IDF (Term Frequency-Inverse Document Frequency) weighting suggested by Jones (1972). The major limitation of the raw frequency counts is that TF-IDF down-weights terms that occur frequently across many documents, hence containing minimal discriminative information about the documents. Ramos (2003) established that TF-IDF out-classes the raw frequency representations in document classification problems and this has been concurred by numerous sentiment analysis studies. The use of bigrams to the TF-IDF representation as used in this project was confirmed by Pang et al. (2002) as they demonstrated bigrams to be useful in sentiment classification because of the ability to capture phrase-level sentiment expressions like not good and highly recommend.

3.5.2 Word Embeddings and Contextualised Representations

Literature more recently had been changed to dense vector representations. Mikolov et al. (2013) released Word2Vec which showed that word embedding neural nets stored semantic connection of words as continuous vectors, and was much more effective on semantic chores compared to sparse TF-IDF representations. Later Devlin et al. (2019) proposed BERT (Bidirectional Encoder Representations from Transformers), a pre-trained transformer model generating contextualised representations, to whom the same word is presented with a different vector representation under different circumstances. BERT and its variants set state-of-the-art performance on practically all NLP tasks, including sentiment analysis when Sun et al. (2019) evaluated fine-tuned BERT receiving over 95% accuracy on tasks in binary sentiment classification.

The choice against adopting BERT in the project was made with respect to the limitations of computational resources and the aim to make a classical baseline reproducible in the first place. These developed methods are however explained as future directions in Section 12.

3.6 NLP Preprocessing in Sentiment Analysis

Preprocessing has been studied as being an essential part of NLP pipelines. Haddi, Liu, and Shi (2013) have shown that systematic, pre-processing of text i.e. stopword removal, negation processing and stemming/lemmatisation enhanced the accuracy or the SVM-based sentiment classification, up to 8.1 percent relative to unprocessed baselines. Importantly, in their research they discovered that sentiment tasks were adversely affected by removing all stopwords in an indiscriminate manner since such negation words like not, never and cannot

essentially invert the polarity of the other adjacent words. This observation was the direct motivating factor that led to the negation-saving stopword deletion policy achieved in the project.

Jia, Yu and Meng (2009) reviewed the role of negation handling on sentiment classification and observed that the scope of negation detection even in its simplest state of preserving negation words delivered the same result of improving the accuracy of classification across various datasets with varied classifiers. Saif et al. (2014) also discussed the effects of stopword removal techniques on sentiment analysis in Twitter and the study demonstrated that task based stopword list is always the best compared to the generic list.

3.7 Class Imbalance in Sentiment Datasets

Imbalanced classes phenomenon is widespread when working with sentiment analysis data because e-commerce platform visitors are more likely to leave reviews when feeling strongly positive, strongly negative, making the clustering of the Neutral class sparse. A systematic study of the effects of class imbalance on neural network classifiers by Buda, Maki, and Mazurowski (2018) established that relatively large proportions of class imbalance can have a notable impact on the classification of each majority class. To address this, Chawla et al. (2002) introduced SMOTE (Synthetic Minority Over-Sampling Technique), but simpler random resampling approaches to the problem enhance a classical machine learning classifier, as used in this project, have been demonstrated to be equally as effective (Japkowicz and Stephen, 2002).

3.8 Summary

It is already known in the literature that the classical methods of machine learning, specifically Logistic Regression and SVM in combination with TF-IDF features representations, can still be considered a competitive baseline to sentiment classification, despite the recent dominance of transformer-based models. Preprocessing choices, in particular, negation handling, do affect model performance. Three-class sentiment analysis has been repeatedly stated to be more difficult to classify than binary classification, and Neutral class has been identified as the main error source of misclassification in all the approaches. These lessons were used as guidance behind all of the design choices in this project and they provide the theoretical basis that the methodology in the next section describes.

4. Methodology

4.1 Justification for CRISP-DM

The **CRISP-DM (Cross-Industry Standard Process for Data Mining)** methodology was the general framework that was used in this project. CRISP-DM was chosen as compared to other methodologies including SEMMA, KDD, mainly due to three major reasons. To begin with, CRISP-DM is clearly business-oriented, and the practitioner must establish and reach back to business goals on a consistent basis as well technical decisions should be meaningful and results-focused instead of scholarly abstract. Second, its non-linear, iterative design allows the exploratory character of NLP projects, with discoveries made in the preparation phase often making it necessary to go back to previous stages. Third, CRISP-DM is the most common data mining methodology widely used in practice in the industry and in research projects, which offers a standardised, reproducible methodology of data mining making the rigorous documentation and peer evaluation possible.

SEMMA (Sample, Explore, Modify, Model, Assess) is pragmatic but too tool-focused and does not have the business understanding phase which is an important phase of this project. KDD (Knowledge Discovery in Databases) is a comprehensive framework that, nevertheless, is database-oriented and fails to take into consideration deployment concerns, which is mandatory in the scope of the work that this project is required to conduct.

4.2 CRISP-DM Framework Overview

The six steps used in CRISP-DM in relation to this project are outlined below:

Phase	CRISP-DM Step	Project Application
1	Business Understanding	Define sentiment classification objectives and success metrics
2	Data Understanding	Load, explore, and analyse the Amazon Fine Food Reviews dataset
3	Data Preparation	NLP preprocessing pipeline, balancing, TF-IDF feature extraction
4	Modelling	Implement and tune Naive Bayes, Logistic Regression, and SVM
5	Evaluation	Compare models using F1-score, precision, recall, confusion matrices
6	Deployment	Build real-time sentiment prediction function for unseen reviews

4.3 Phase 1- Business Understanding

The business issue was clearly detailed in the form of the inability of e-commerce organisations to receive and respond to massive amounts of customer feedback at the scale. This target task was set to three-class sentiment classification Negative (1-2 stars), Neutral (3 stars), and Positive (4-5 stars) where it was known since star ratings would not be included, in order to create the inference as similar to real world conditions as possible where numeric ratings could be absent or unreliable.

The weighted F1-score of ≥ 0.80 was chosen as the primary success metric as, unlike accuracy, the dataset even after balancing is not a trivial three-class classification task where the disparities in per-class performance are known to occur and should be taken into consideration. The weighted F1-score provides the balance between accuracy and recall in every class depending on its support, therefore being more informative than accuracy on multi-class problems.

4.4 Phase 2- Data Understanding

The Amazon Fine Food Reviews data was loaded and was undergone to a thorough exploratory data analysis (EDA). The most important tasks were to analyse the raw form and column layout of the data, counting the missing values, studying the distribution of the star rating, code ratings to sentiment categories, and the visualisation of the imbalance of the classes. Length distributions of reviews were calculated and compared by their sentiment type to determine the presence of viable feature engineering and make preprocessing choices. Section 5 (Dataset Description & Exploration) corresponds to the findings of this phase, with a written account of the same.

4.5 Phase 3 - Data Preparation

The most technically intensive part of the project was data preparation and it comprised three significant sub-processes:

3a Class Balancing: The raw data had extreme positive skew with 78.2% of the reviews having 4-5 stars. Random resampling with replacement to balance minor classes helped to avert bias by majority classes in the dataset; therefore, each minority class was resampled to 20,000 reviews in total (60,000 in total) The reason behind this threshold was to optimise the size of the dataset and still allow hyperparameter tuning to be computationally feasible.

3b NLP Preprocessing Pipeline: It used a pipeline consisting of eight sequential text preprocessing steps, including: HTML tag removal, URL and email stripping, contraction expansion, special character removal, lowercasing, word level tokenisation, negation-preserving stopword removal and WordNet lemmatisation. All the steps were supported by both literature and empirical data of NLP studies. This pipeline has been fully documented in Section 6.

3c - Feature Extraction: TF-IDF vectorisation between a range of unigram-bigram with 15000 maximum features, sublinear term frequency scaling, minimum document frequency 3 and maximum document frequency of 0.90 was done. The vectoriser was fitted on the

training set (80, 48,000 reviews) to avoid data leakage, and the same was applied to both training and test sets.

4.6 Phase 4 - Modelling

Three classical supervised machine learning algorithms have been chosen and run:

Multinomial Naive Bayes: It was chosen as a probabilistic base classifier, which is theoretically based on the Bayes theorem and on the assumption of conditional independence. It was the natural choice in determining a lower bound in performance since it is already known to be effective with sparse TF-IDF matrices, with minimal computation.

The **Logistic Regression** (multinomial, L-BFGS solver) was chosen as the main linear classifier because it reportedly performs well on high-dimensional text classification problems as well as because it can yield calibrated probability estimates over more than two classes, which is useful when the overall high-metabolic costs of applying the estimator can be borne by downstream applications of the resulting predictions like routing based on confidence.

Linear Support Vector machine (LinearSVC) was chosen because of its theoretical appropriateness with high dimensional sparse feature spaces in which the linear kernel is not only computationally efficient but also always effective. The maximisation of margin that SVM uses renders it especially resistant to the high-dimensionality of TF-IDF representations.

Each of the three models was optimised using hyperparameter optimiser, **GridSearchCV** with **3-Fold Stratified Cross-Validation**, to ensure that the choice of a parameter was made based on generalisation performance, rather than on training set performance, and that the proportion of classes in all folds was the same.

4.7 Phase 5 — Evaluation

A held-out test set (peering at 20 percent of the balanced set, 12,000 reviews) was used in model evaluation by stratification to ensure the proportions of classes were the same. The measures of evaluation were accuracy, macro-averaged F1-score, weighted F1-score, weighted precision, weighted recall, per-class classification report, and normalised confusion matrix, as well as a heatmap of per-class F1-score. Reporting both macro and weighted F1-scores is an intentional decision, since the former is fair to the classes and the latter is responsive to the underperformance of the Neutral class, whereas the former measures all the classes equally, whereas the latter considers the support of the classes.

4.8 Phase 6 — Deployment

The best model that performed well Logistic Regression was operationalised in the deployment stage in a real time sentiment predictor. This can take any text that is raw in the English language as input, run the entire preprocessing pipeline and transformation of TF-IDF and give a predicted sentiment class with the interpretable label. This indicates that the pipeline is practically able to be hooked into a production system (customer feedback dashboards, automated review routing engines, or platform of product quality monitoring, etc).

4.9 Methodology Summary

The CRISP-DM framework provided a principled, iterative structure that ensured every phase of this project was purposefully connected to the defined business objective. The methodology facilitated systematic documentation of design decisions, rigorous experimental evaluation, and critical reflection on results — all of which are essential to producing a reproducible and academically credible data mining study.

5. Dataset Description & Exploration

5.1 Dataset Overview

The dataset, used in this project, is **Amazon Fine Food Reviews**, which is a corpus publicly available on Kaggle and firstly collected by McAuley and Leskovec (2013) in a large-scale investigation into the interaction between quantity of reviews and quality and the perception of a product. The data will include consumer reviews on fine food products posted on the Amazon online shopping platform that covers a time spanning **October 1999 to October 2012** or more than a decade of unedited consumer sentiment in the food and grocery sector.

The raw data consists of **568,454 reviews** in **10 columns**, the most useful of which to this project being the **Score** column which is a star rating of the reviewer out of 1 to 5 and the **Text** column which is the complete free-text body of the review. Other columns such as **Summary**, **ProfileName**, **HelpfulnessNumerator**, and **HelpfulnessDenominator** were not modelled because the aim of the project is to classify texts in form of text only sentiment classification. Dropping rows that either had a value missing in the **Score or Text fields**, the usable dataset consisted of **568, 427 reviews**.

The dataset I used:

https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews?utm

5.2 Sentiment Label Mapping

The following schema was used to map star ratings into three classes of sentiment based on the normal practice of the literature on sentiment analysis:

Star Rating	**Sentiment Class**	**Label**	**Justification**
1–2 stars	Negative	0	Definitely dissatisfied with reviewer; language is mostly critical.
3 stars	Neutral	1	Undecided or undiscriminating opinion; language makes no way out or is neutral.

Star Rating	Sentiment Class	Label	Justification
4–5 stars	Positive	2	satisfied or much enthusiastic; words are mostly positive.

This mapping concurs with the three-class formulation that Agarwal et al. (2011) made and it is highly established in product review sentiment analysis studies. The three-star boundary was carefully considered as a separate Neutral category instead of having it merged together with two extremes, which would reduce a true uncertainty example lab review and would cause artificial noise in the training example.

5.3 Class Distribution Analysis

A look at the raw star rating distribution showed that the star rating is heavily skewed towards a positive and is a typical feature of e-commerce review data and can be explained by the self-selection behaviour of satisfied customers, who more often than not review satisfied products compared to moderately unsatisfied customers, and a small but vocal minority among the most unsatisfied.

The raw sentiment distribution was as below:

Sentiment Class	Raw Count	Percentage
Positive (4–5 stars)	443,777	78.1%
Negative (1–2 stars)	82,010	14.4%
Neutral (3 stars)	42,640	7.5%

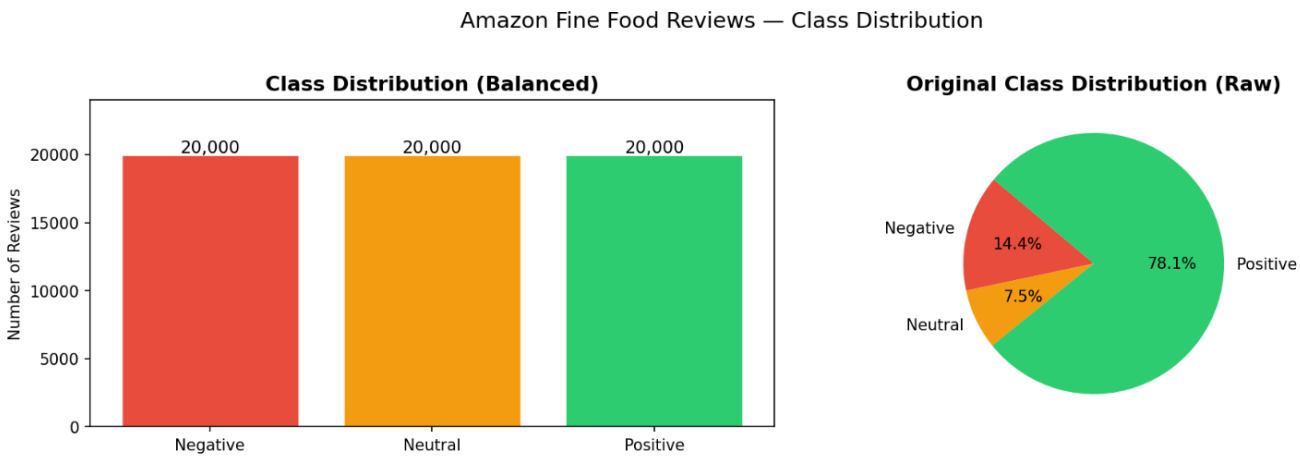

Figure 1: Class distribution of the Amazon Fine Food Reviews dataset showing balanced and original distributions.

Class Distribution Balanced Dataset (left) and Original Raw Distribution (right). The pie chart validates the deplorable 78.1% Positive skew in the raw data, whereas the bar chart validates ideal balance at 20,000 reviews per class post resampling.

Such a harsh imbalance, in which the Positive reviews covered 78.1% of the total corpus and the Neutral reviews comprised the humble 7.5% according to Figure 1 (right) was a grave danger of majority class prejudice. A learner with an imbalanced classification based on this raw distribution would be strongly encouraged to follow the Positive class and this would show a very high rate of accuracy but would not perform the Negative and Neutral classes well. This is what makes the accuracy a poor measurement criterion of such an undertaking.

In order to remove this bias the dataset was resampled to 20,000 reviews of each class (60,000 in total) and after balancing all three sentiment classes the bar distribution is perfectly uniform as is shown in Figure 1 (left). The minority Neutral class is randomly oversampled and replaced and the majority Positive and Negative ones are random undersampled. Such a balanced distribution, in turn, made sure that both further models were trained in equal conditions of fairness and without any bias that favors one of the classes over another.

5.4 Review Length Analysis

Review length: The count of words in a review was analysed per review in all three classes of sentiments, having balanced them to determine distributions of data in respect to preprocessing and feature extraction decisions. Figure 2 shows the length distribution of reviews per of the three classes, each having the x-axis bounded at 400 words to be visualized but any farther and it is thought to reflect a small percentage of the corpus.

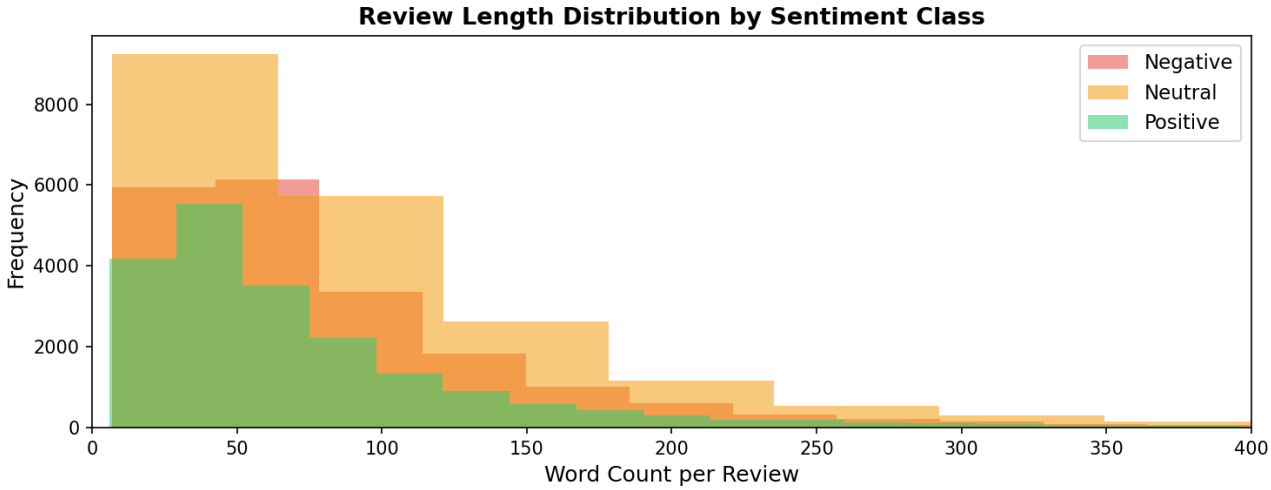

Figure 2: Review Length Distribution by Sentiment Class (word count per review, capped at 400 words).

One can make a couple of significant observations based on Figure 2 directly:

Right Skewness Distribution: The sentiment classes are all skewed to the right giving rise to a large region of 0-100 words and then trending rapidly to the right. This can be attributed to the character of product reviews, most consumers give their review in brief and not in length.

Neutral Reviews Have Longest Tails: The longest distributional tail represents the Neutral group (orange), and continues far beyond 300 words, where the frequency is significant, - way farther than the Negative and Positive groups. It is a linguistically significant result: the higher a reviewer in the three-star range rates a product, the higher the possibility of elaborating on the positive and negative features of a product leading to longer and balanced writing. This long-length is also somewhat the reason why the most difficult reviews to categorize are Neutral ones because of their higher numbers of mixed contents, they bring more ambiguity to the feature space.

Shortest Positive Reviews: Positive class (green) decreases the most, and most of the reviews are within the 0-50 word range. This is indicative of the nature of highly satisfied consumer behavior of making short, glorifying recommendations but not elaborate ones, a phenomenon that is consistent with the psychology of positive consumer review.

Negative Reviews Are Mid: Negative as it (red) occupies an intermediate place, being longer than Positive review on average, however being shorter than Neutral review at the distributional tail. Unsatisfied consumers will sometimes elaborate on their complaints in moderately terms, and the linguistic emphasis of negative remarks on particular complaints, leads to more focussed, moderately lengthy writing as opposed to the moderation of positive remarks typical of the impartially-minded Neutral reviewers.

These properties of length directly suggest feature extraction: the dominance of short-to-medium length reviews is confirmed that word-level TF-IDF is a reasonable feature representation, whereas character-level or subword tokenisation plans (as in transformer models) are not needed since the reviews are largely within manageable length range.

5.5 Dataset Quality Assessment

A structured quality evaluation of the data was carried out according to the following dimensions:

Missing Values: Once the null entries in the Score and Text columns were eliminated, there were no values in the remaining columns that were considered missing in the dataset, which predetermined the absence of unwanted values in the input of the preprocessing pipeline.

Duplicate Reviews: It is known that the Amazon Fine Food Reviews dataset has a low percentage of duplicate or near-duplicate reviews, which can be explained by automated syndication of product variants via reviews. Although in this project no formal deduplication step was taken, the strategy of the such random resampling and stratified train-test split will help reduce the chances of having the same reviews being included in both the training and test sets.

Encoding Problems: The raw data included some non-ASCII encoder characters, HTML entity characters and encoding artefacts added to it by the original data encoding process.

These were systematically handled in the preprocessing pipeline with the removal of HTML tags and special character normalisation which is described in detail in Section 6.

Language: The scope of the project restricts all the reviews to the English language. There was no need to have multilingual filtering.

5.6 Summary

The Amazon Fine food Reviews data is a good corpus, large-scale, and domain-specific and it is rich to carry out sentiment analysis studies. The EDA performed herein has identified two salient properties directly influencing the methodology of the project: the critical class imbalance that demanded the intervention of resampling, and the length distribution of the reviews that have been performed in a language-differentiated classification pattern with the Neutral reviews as constant and longer length-wise, and more linguistically advanced than the Positive and Negative ones. Such results highlight the reason behind Neutral class being the most difficult to classify - a trend that would be demonstrated with the model assessment findings in Section 8.

6. NLP Preprocessing Pipeline

6.1 Overview

The effectiveness of the preprocessing stage of the NLP pipeline is significant in determining the overall quality of the pipeline. Raw text data - especially that generated by users like product reviews are noisy, inconsistent, and informally structured by nature. It holds HTML artefacts, URLs, contractions, punctuation, numbers, and domain terminology that needs to be procedurally converted into a clean and normalised machine usable form prior to any meaningful feature extraction taking place. This part records every stage in the eight stages long preprocessing pipeline applied in this project and their justifications based on the literature of NLP as well as their justification through empirical results of the dataset.

All the 60,000 balanced reviews were uniformly applied to the pipeline after which feature extraction and model training was carried out.

6.2 Preprocessing Pipeline — Step-by-Step Documentation

Step 1 - HTML Tag Removal

The data provided in the Amazon Fine Food Reviews was gathered through web scraping and includes leftover HTML markups in the middle of reviews like <br>, <br>, and

<a href=...>These labels do not have any semantic meaning that concerns sentiments.

classification and would put spurious tokens in the feature space otherwise. The all-HTML tags were removed using a regular expression pattern (<[^>]+>)All HTML tags were removed, and the words were surrounded by white space to maintain the boundaries of words.

Step 2 - URL and Email Address Removal

The reviewers sometimes incorporated links in the form of URLS (e.g. the product links, reference sites) and email addresses. A semantically irrelevant part to sentiment is these strings and, held, would produce high-cardinality unique feature space tokens unhelpful to prediction. Regular expression pattern on strings of **http**, **www**, and **@**- strings was used to eliminate the same.

Step 3 - Contraction Expansion

Contracted forms to a large extent are used in informal review writing like, don't, won't, isn't, can't. Unexpanded contractions allow a form of ambiguity that is called tokenisation ambiguity the word don't can be tokenised by the fusion of a single opaque token instead of the semantically meaningful two-token token sequence do not. There was a custom contraction dictionary used to propagate all standard English contractions before further processing to make sure that the negation words have been achieved well as independent tokens to be preserved downstream.

Step 4 - Special Character and Number Removal

All characters, except those that were parts of a word (such as punctuation and digits and currency symbols and emoticon) were eliminated by a character-class regular expression ([^a-zA-Z\s]). Although the use of emoticons can be sentiment based in the case of social media, they have a relative limited number of emoticons in the Amazon Fine Food Reviews corpus and they are inconsistently represented throughout the dataset, which makes them unreliable features. The numbers were also eliminated because they do not have any generalised sentiment connotation in the domain of food reviews.

Step 5 – Lowercasing

Text was reduced to lower case to allow semantically similar words used in different forms like Good, good, and GOOD to be counted as a single count. It is a typical and equally universal normalisation step in text classification pipelines.

Step 6 - Word-Level Tokenisation

NLTK word tokenization function (word_tokenize) was used to perform tokenisation based on the Penn Treebank tokenisation standard with NLTK word tokenize. Word level tokenisation was chosen over the option of either character or subword level tokenisation due to the analysis based on review length that was performed in Section 5 that established that most of the reviews have reviewed lengths that are between 0-150 words a range that is computational and semantically sufficient in the word level representation.

Step 7 - Negation-Preserving Stopword Removal

The usual preprocessing involves stopword removal, which removes high frequency, low information, function words, including the, is, at and which that add minimal discriminative information to sentiment classification. The NLTK English stopword list has been used, but a critical modification was made: **the words of negation were not included in the stopword list, but rather removed and not turned into tokens**.

The negation words that were preserved were: no, not, nor, neither, never, nobody, nothing, nowhere, cannot, don't, won't, isn't, aren't, wasn't, weren't, hasn't, haven't, didn't, doesn't, wouldn't, couldn't, shouldn't, very, too.

This choice is strongly supported by Figure 4, where it can be seen that it is not the words that most often appear in all three classes of sentiment, it is not, appearing more than 35,000 times in Negative class, and more than 36,000 in Neutral, and almost 19,000 in Positive. The frequency of not among all classes proves that it is not a noise token, on the contrary, it is an important sentence modifier. Blind deletion of not would have a fundamental effect of polluting the semantic polarity of the phrases like not good, not worth and not recommended and may end up reversing the meaning of such phrases completely. This observation in Figure 4 empirically supports the negation preservation strategy in direct proportion.

Also, tokens that contained two characters or less were eliminated because they have minimal semantic meaning when used in food review.

Step 8 - WordNet Lemmatisation

The final normalisation step which was chosen was lemmatisation instead of stemming. Based on heuristic suffix-stripping rules (e.g. turning tasty into tasti, flavours into flavor), stemming algorithms like Porter or Snowball break down a word down to a dictionary stem (e.g. a word). Lemmatisation, in its turn, applies the morphological dictionary (WordNet) to simplify the words to the canonical version in the dictionary base - both readability and semantic soundness are maintained. As an illustration, the word tasting, tasted, and tastes are all rightly diminished to taste. This is necessary especially in the food review field whereby sensory words like tasted; tastes; tasting and taste are recommended to be grouped as one feature. NLTK was used to perform wordnet lemmatisation with WordNetLemmatizer which uses default noun-based lemmatisation.

6.3 Preprocessing - Before and After Example

The following example shows how a raw review can be transformed over time, i.e. the cumulative effect of the pipeline:

BEFORE (Raw): This is absolutely the best granola I have ever tasted. Will buy again!

AFTER (Preprocessed): absolutely the best granola ever taste buy

The pipeline was able to remove stopwords, lowercase all tokens, lemmatise the word tasted to taste, and retain only content words that had semantic meaning to them - reducing the review by 13 tokens to 6 very much informative tokens.

6.4 Post-Preprocessing Text Analysis

After preprocessing, the word frequency and word cloud visualisation were applied to the cleaned corpus in order to confirm the pipeline worked effective, and a discriminative vocabulary pattern of different sentiment categories was obtained.

6.4.1 Word Cloud Analysis

Figure 3: Word Clouds by Sentiment Class

Figure 3 shows word clouds resulting out of post-preprocessed text of each sentiment class with word size dependent on frequency in that sentiment group.

Negative Reviews (red): The prevailing keywords are as follows: product, taste, food, flavor, coffee, and make. It is interesting to note that the most often used word is product, the largest word is used, which is the confirmation that the negative reviewers often talk about the product itself, but not in particular qualities and the quality of the product in general. Preeminence of food, coffee, and flavor contains signs of domain-specific language of food-review corpus.

Neutral Reviews (orange): The most prevailing words are good, taste, coffee, flavor and product. The widespread occurrence of the word good in the Neutral group is a very insightful observation - it shows that the neutral reviewers are using positive language, yet are simultaneously going on to say negative things about the same phenomenon, meaning that the neutral group and the Positive one are lexically close and the difficulty of the classification.

Positive Reviews (green): The frequently used words are taste, great, good, flavor, coffee, love and store. It is also evident that the Positive class stands out clearly against the other two because of the high evaluative nature of the terms great, love, and best, which are generally features of passionate approval.

6.4.2 Top 15 Most Frequent Words Analysis

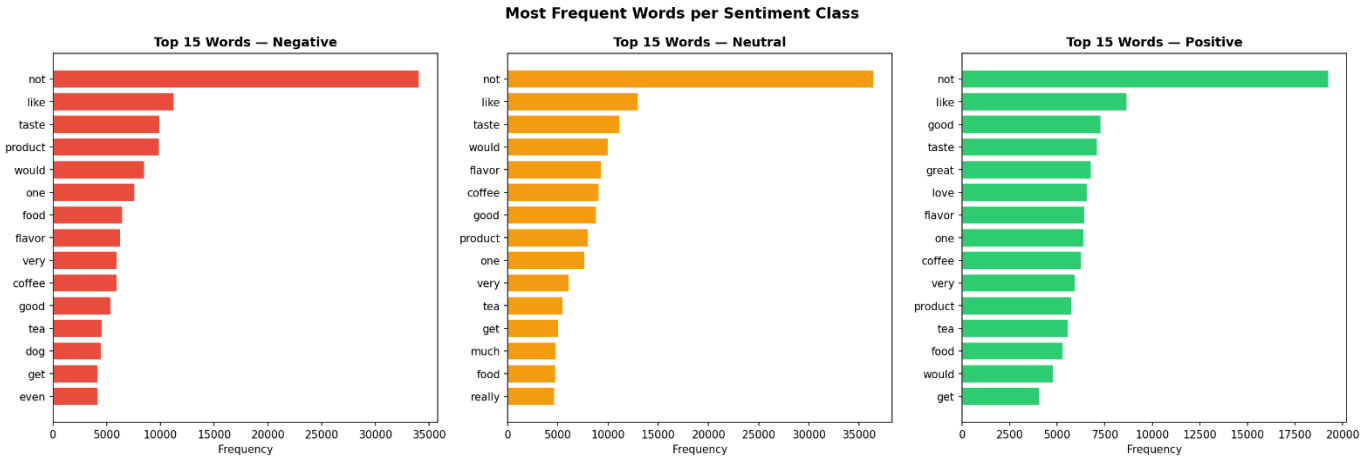

Figure 4: Top 15 Most Frequent Words per Sentiment Class

Figure 4 gives a quantitative follow-up of the word clouds by showing the 15 most frequent words in each of the classes as horizontal bar graphs with frequencies.

Figure 4 presents the most important observation that **not is a high-frequency word in all three sentiment categories** with the frequencies of about 35,000 (Negative), 36,000 (Neutral), and 19,000 (Positive). This simply confirms the choice of negation preservation design – **not** only does it occur, but is the only dominant token in the whole corpus, and its deletion would have been a drastic and deliberate eradication of sentiment-carrying data.

There is much overlap between the top word lists in all three classes, in addition to not, the words like, taste, would, flavor, coffee, one, very, product, tea and food are found in the top 15 of several classes. This lexical confusion can be used as an important explanatory variable behind the classification challenge witnessed in model evaluation, specifically in the case of the Neutral class: when the most common words are shared amongst classes, the classifier has to use a smaller, finer grain of distributional differences, like the interaction of not with good versus not with great, in order to make accurate predictions. This observation also supports the merit of the **bigram TF-IDF representation** that is used in the feature extraction step since bigrams can encode such important word-pair relations.

6.5 Summary

The preprocessing pipeline is an eight-stage publication of the process that systematically converted raw and messy Amazon review texts to clean and normalised token sequence, usable in extracting TF-IDF features. The most impactful design actions negation word preservation and lemmatisation, but not stemming, were shown to be empirically supported through the post-preprocessing analysis in Figures 3 and 4. The term frequency analysis showed that the lexical overlap had been made significant among the emotion classes especially between the Neutral and Positive reviews, and this gave a good indication of what was to be expected during the modelling stage where the Neutral will pose the most challenging classification problem.

7. Feature Extraction Methods

7.1 Overview

After the preprocessing pipeline mentioned in Section 6, the text has to be cleaned and in the form of a number that machine learning algorithms can operate with. It is this change - the feature extraction or text vectorisation - that is one of the most material decisions in any NLP pipeline, since the representation decision will dictate precisely what the classifier will get access to and what it will be wiped out. The following part reports and defends the approach to feature extraction that may be applied in this project.

7.2 From Text to Numbers - The Core Challenge

Machine learning algorithms do not process individual text, but work on numerical matrices. The basic issue of text feature extraction is then to encode the semantic and statistical characteristics of language into a fixed length numeral list whilst maintaining as much of the discriminative information as is possible. There are several methods of such transformation, starting with the simplest usage of frequencies to the complicated neural embedding spaces. The proposed project uses **TF-IDF with unigram and bigram features** of a project that occupies that just right zone between simplicity and expressiveness of classical sentiment classification tasks.

7.3 Bag-of-Words- Conceptual Foundation

The conceptual framework that TF-IDF is based on is the **Bag-of-Words (BoW)** model. Under the BoW style, a document is represented as a word count matrix of the number of occurrences of words in a fixed vocabulary, and represents all information about grammatical procedure or word order. Although this is a semantically sparse representation (the food was not good and the food was good are virtually the same BoW vectors) it has been surprisingly effective at scale, at document level sentiment classification, where the distribution of words over a review is highly predictive of the sentiment classification.

The main weakness of raw BoW count vectors is that very common words, e.g. food, product, taste in this corpus, will tend to occupy most of the feature space merely due to their frequency (not due to being discriminative of anything). The TF-IDF deals with this shortcoming directly.

7.4 TF-IDF Vectorisation

TF-IDF (Term Frequency-Inverse Document Frequency) is a statistical weighting scheme that adjusts raw word frequencies by their informativeness across the document corpus. The TF-IDF weight of a term $t$in document $d$is computed.

$$TF - IDF(t, d) = TF(t, d) \times \log\left(\frac{N}{df_t}\right)$$

Where:

- **TF(t, d)** — Term Frequency: how often term $t$appears in document d

- **IDF(t)** — Inverse Document Frequency: $\log\left(\frac{N}{df_t}\right)$, where $N$is the total number of documents and $df_t$is the number of documents containing term t

Common terms to a particular document and uncommon terms to the corpus are those with a high TF-IDF, which is an indicator of a strong local relevance and a low global commonality of a discriminative feature. In contrast, words occurring in almost all of the documents have virtually zero IDF weights, which in effect prevents their contribution to the feature space.

7.5 Implementation Parameters and Justification

Experimenting with TF-IDF TFidfVectorizer TFidfvectorizer in scikit-learn was configured in the following parameter values, which were chosen deliberately and explained:

Parameter	**Value**	**Justification**
max_features	15,000	Regularly retains the 15,000 most informative words - tradeoffs between feature expressiveness and computationally efficiency and overfitting of uncommon words.
ngram_range	(1,2)	Supports both single-word and multi-word phrases e.g. bigrams - allows the model to encode phrases with sentiment.
sublinear_tf	True	Applies log(1 + tf) scaling to term frequencies, reducing the dominance of very high-frequency terms and producing a more balanced feature distribution
min_df	3	Eliminates words that are found in less than 3 documents - gives rid of token words which are probably noise.
max_df	0.90	Suppresses terms that occur in at least 90% of the documents - inhibits terms that are nearly universal and unselective.

Vectoriser was configured only on the training set (48,000 reviews) and further used to transform the training set and test set. This is a very strict protocol which makes sure that there is no leaking of data - the vocabulary and the IDF weights are calculated only with the help of the training data, and the test set is truly an unseen evaluation case.

7.6 Justification for Bigram Features

The minimum impactful feature engineering choice in this project, which was directly inspired by the post-preprocessing analysis in Section 6, was the addition of bigrams (two-word sequences) to the TF-IDF representation. The problematic observation of Figure 4, namely the fact that the token of not is the most frequent of all three sentiment classes is the frequency of that word, which generates a particular and familiar problem with the tool of unigram representations: the word **not** by itself does not have a signal that belongs to a particular class, since it appears too high frequency in Negative, Neutral, and Positive reviews.

The ambiguity is overcome by bigram features which encode the contextual meaning of the word not. For example:

Bigram	**Sentiment Signal**
not good	Negative
not bad	Positive (implied)
not worth	Negative
highly recommend	Positive
not recommend	Negative
not sure	Neutral

These features at the phrase level cannot be represented in a unigram representation but can be encoded at once as bigram features, giving classifier more discriminative information which is rich with more context. This design rationale has been justified by Pang et al. (2002) who established good accuracy upgrading as a result of bigram addition in sentiment assignment duties.

7.7 Resulting Feature Matrix

After transforming to TF-IDF, the data took the form of a sparse numerical matrix whose dimensions were as follows:

Split	**Documents**	**Features**
Training set	48,000	15,000
Test set	12,000	15,000

The resulting matrix is **sparse** - the enormous percentage of non-zero entries is generated by an exceptionally small percentage of the 15,000-term in the review of a single individual. All of these were implemented in scikit-learn using the sparse matrix implementation (CSR format) which provides memory-efficient storage and computation even when using a high-dimensional feature space.

7.8 Why Not Word Embeddings?

Although dense word embedding representations (Word2Vec, GloVe, BERT) provide a more detailed semantic encoding whenever compared to TF-IDF, they have not been used in this project due to the following valid reasons:

Computational constraints: It is expensive to fine-tune or train transformer-based models like BERT on 60,000 reviews on a single graphics card and significantly time-intensive than can be practiced within the boundaries of this project.

Baseline establishment: TF-IDF with classical machine learning classifiers: The classical, reproducible baseline in sentiment analysis studies. Having a good base is a prerequisite of meaningful comparison with advanced techniques.

Compatibility: TF-IDF sparse matrices are intrinsically compatible with three classical classifiers that were used in this project, that is, Naive Bayes, Logistic Regression, and SVM, without any architectural changes.

The development of word embeddings and transformer-based representations is mentioned as an essential way of going ahead in Section 13.

7.9 Summary

On the basis of its good empirical history in sentiment classification, its direct usability with classical machine learning classifiers, and its capability to encode critical sentiment bearing phrases using bigram features, TF-IDF vectorisation with unigram and bigram features and 15,000 maximum features was chosen as the feature extraction method that would be used in this project. The train-fit, train-transform, test-transform protocol was very strict and the leakage-free use of an experimental design. The respective sparse feature matrices were 15000-dimensional came out as the input to the three classification models reported in Section 8.

8. Algorithm Implementation and Comparison

8.1 Overview

Here, the theoretical basis, implementation process, hyperparameter optimisation process, and cross-validation outcomes of each of the three takes used in this project are reported. The models were all trained on the feature matrix of TF-IDF in Section 7 which comprises 48,000 training samples and 15,000 features. This was done by choosing hyperparameters via GridSearchCV with 3-Fold Stratified Cross-Validation to make sure that any performance estimates reported are obtained by generalisation to data not seen during training but actual performance of the model.

Three algorithms were chosen as they represented three fundamental different learning paradigms Multinomial Naive Bayes (probabilistic), Logistic Regression (discriminative linear), and Support Vector machine - LinearSVC (margin-based). Collectively, each of these three models offers a full classical machine learning base on the three-class sentiment classification problem.

8.2 Model 1- Multinomial Naive Bayes

8.2.1 Theoretical Foundation

Multinomial Naive Bayes (MNB) is an example of a probabilistic generative classifier based on the Bayes theorem. It is a model of the posterior probability of a given class C_k

that a document d belongs to, the prior probability of the class and the likelihood of observing each feature respectively and the given class. The rule of classification is as follows:

$$P\left(\frac{C_k}{d}\right) \propto P(C_k) \times \prod p\left(\frac{t_i}{C_k}\right)$$

Where $\mathbf{P(C_k)}$ is the prior probability of class k, which is estimated by the frequency of the training data in classes, and $\mathbf{P(t_i \mid C_k)}$ is the estimated value, based on smoothed maximum likelihood using the training data, of conditional probability of term t_i

appearing in a document of class k. The naive assumption is that, the conditionally independent features given the label are all independent regardless of the label.

Though this assumption is almost never true in the face of natural language (where words co-exist in patterned patterns) Multinomial Naive Bayes has been shown to perform well in empirical terms over and over again in text classification tasks, especially when using TF-IDF weighted features (Rennie et al., 2003). Multinomial variant is uniquely positioned to take advantage of discrete count or frequency features thus making it theoretically ideal in the sparse TF-IDF representation of this project. Its main practical strengths include it being a computationally efficient algorithm, interpretable, and can be trained quickly, and as such, is a good baseline classifier when trying to set a lower limit on the performance of any given classification.

8.2.2 Hyperparameter Tuning

The Laplace smoothing parameter α (alpha) is the critical hyperparameter of Multinomial Naive Bayes as it is used to avoid zero-probability estimates of those words that appear in the test set but do not appear in the training examples of a given class. Any unseen terms would then make the entire posterior probability collapse to zero with no way of dealing with the unknown vocabulary that the classifier has not encountered in the training.

The candidate values assessed by the grid search CV include:

α € {0.01, 0.10, 0.50, 1.00}

As we can see somewhat in Figure 5 (left panel), the F1-Weighted score of cross-validation continued to be nearly perfectly flat in the whole range of alpha values and leveled off at around 0.70 in spite of the smoothing strength used. Such almost perfect insensitivity to alpha is a familiar behavior of Multinomial Naive Bayes when working on large and balanced data-sets indices, i.e. when one has enough training examples per class, the additive smoothing correction becomes insignificant in proportion to the posterior set of probability estimates.

The best value was taken as alpha = 1.0 that would be taken as standard smoothing in Laplace (add-one). This value yielded the best cross-validation F1-Weighted value and it represents the theoretically traditional default, which is that every unseen term has an equal pseudo-count weight.

8.2.3 Implementation Details

The Multinomial Naive Bayes model was optimized and tested as in the following:

Parameter	Value
Model	MultinomialNB(alpha = 1.0)
Training input	TF-IDF sparse matrix (48,000 × 15,000)
CV strategy	Stratified K-Fold (k = 3)
Scoring metric	f1_weighted
Best CV F1-Weighted	~0.70
Tuned parameter	alpha = 1.0

Naive Bayes despite its simplicity had a reasonable baseline performance. Its main weakness in this activity is the assumption of conditional independence: sentiment is often carried by combinations of words, as in not good and highly recommend, not single words isolated - co-dependencies which MNB cannot in its structure model. This drawback is directly translated in its low test F1-Weighted Value of 0.7106 as opposed to the other two models.

8.3 Model 2- Logistic Regression

8.3.1 Theoretical Foundation

Logistic Regression (LR) is a linear discriminative model of the posterior distribution of the default class, using input features, and not assuming the underlying distribution of features - unlike the generative approach of Naive Bayes. In the case of the three-class sentiment task of this project, the multinomial (softmax) formulation is applied, where the likelihood of class k is subsequently computed as:

$$\mathbf{P(C_k \mid x) = \exp(w_k x) / \Sigma \exp(w_j x)}$$

With x denoting the TF-IDF feature vector of the input document, w_k denoting the weight vector learnt by training class k and K being the number of classes in total. The training of the model involves the minimisation of multinomial cross-entropy (log-loss) on the training set with L-BFGS optimisation algorithm (Limited-memory Broyden-Fletcher-Goldfarb-Shanno) which is an efficient second order optimisation algorithm that is most effective with smooth, high-dimensional optimisation problems.

Logistic Regression does not give the assumption of independence of features as is the case with Naive Bayes. Instead, it optimistically learns a global weight w_i correspond to each TF-IDF feature that directly determines its discriminative value to each class considering correlations between features. **L2 regularisation** is default with a parameter C that is unfixed and the inverse of regularisation strength. The use of a smaller C value provides more substantial regularisation, driving weights to zero. The higher C value will enable the model to fit the training data better but there is a possibility of overfitting. Wang and Manning (2012) have shown that the Logistic Regression with bigram TF-IDF feature always gives a better result as compared to Naive Bayes on longer and more complex text documents, which in this project have been replicated directly.

8.3.2 Hyperparameter Tuning

The regularisation parameter C was the main hyperparameter to tune the Logistic Regression. The parameters that were found by the use of the GridSearchCV were:

C € {0.1, 1.0, 5.0, 10.0}

As in Figure 5 (centre panel), the cross-validation F1-Weighted score went up significantly between C = 0.1 (~0.70) and C = 1.0 (~0.735), implying that the default regularisation strength at lower C was underfitting the high-dimensional TF-IDF feature space. Even more, after C = 1.0, the score decreased slightly at C = 5.0 and C = 10.0, indicating that with greater regularisation more than 1.0, slight overfitting to training data trends not relevant to unseen reviews took place.

The best value found was thus C = 1.0 at which the cross-validation F1-Weighted score achieved the highest value as this is the best trade-off between model flexibility and regularisation in this dataset.

8.3.3 Implementation Details

The ultimate Logistic Regression model was set and tested as:

Parameter	**Value**
Model	LogisticRegression(C = 1.0, multi_class = 'multinomial', solver = 'lbfgs')
Training input	TF-IDF sparse matrix (48,000 × 15,000)
CV strategy	Stratified K-Fold (k = 3)
Scoring metric	f1_weighted
Best CV F1-Weighted	~0.735
Tuned parameter	C = 1.0

The test set F1-Weighted score of **0.7487** in Logistic Regression is the highest F1-Weighted score of any of the three models. This performance-based characteristic, which it assigns different learned weights to each one of the 15,000 TF-IDF features, such as bigrams that are discriminative like **highly recommended** and not worth, is specific to its performance in comparison to Multinomial Naive Bayes.

8.4 Model 3 — Support Vector Machine (LinearSVC)

8.4.1 Theoretical Foundation

Support Vector Machines (SVM) are discriminant classifiers that are used to find the optimal separating hyperplane in the feature space by maximising the distance between the decision boundary and nearest training data points of each class, or, support vectors. The formal SVM optimisation attribute is:

$$\textbf{Minimise: } \tfrac{1}{2}\|\mathbf{w}\|^2 + C \times \Sigma\xi_i \quad \textbf{Subject to: } y_i(\mathbf{w}^T\mathbf{x}_i + \mathbf{b}) \geq 1 - \xi_i$$

Let w be the hyperplane orientation defining the weight vector, b be the bias, C regularisation parameter where margin-error trade-off is considered, and ξ_i slack variables where non-linearly separable data are allowed with yi taking values in be -1 +1. Geometric margin is given as **2 / ||w||**, so maximising the margin is directly equivalent to minimising $||w||^2$.

In the three-class sentiment problem that is to be addressed in this project, the approach used by LinearSVC is the one-vs-rest (OvR) strategy: three independent binary classifiers are being trained: one against each of the sentiment classes and the rest of the classes. Throughout the inference process, every binary classifier generates a decision function score of the input document, and the publication assigned a score of the highest possible label is allocated to the predicted sentiment.

LinearSVC means a linear kernel is adopted, which is theoretically the best when there are sparse features in a high dimensional feature space, as is the case of TF-IDF vectorisation. Joachims (1998) also showed that, where features of text are in high dimensions, the data is usually roughly linearly separable and non-linear kernels like RBF or polynomial kernels are unneeded and wasteful, since their computation essentially requires diversity.

8.4.2 Hyperparameter Tuning

LinearSVC was tuned on the primary hyperparameter of the regularisation parameter C. The candidate values were investigated by the gridsearchcv:

C € {0.1, 0.5, 1.0, 5.0}

Figure 5 (right panel) indicates that the cross-validation F1-Weighted score has an initial value of about 0.727 at C = 0.1, then increased to the highest value of about 0.730 at C = 0.5, after which it tended to decrease steadily at C = 1.0 and C = 5.0 (Figure 5, right panel). This decreasing curve past C = 0.5 shows that increasing the importance of the regularisation term in the SVM causes overfitting of training data noise to high-dimensional threaded TF-IDF space - fitting support vectors to single training examples instead of training a generally applicable decision surface.

The best value chosen had been thus C = 0.5, where the cross-validation F1-Weighted score was the largest, indicating that a moderate to wide margin with a high regularisation is most generalisable on this TF-IDF feature space.

8.4.3 Implementation Details

The Support Vector Machine model that was chosen and tested was as follows:

Parameter	**Value**
Model	LinearSVC(C = 0.5, max_iter = 3000, random_state = 42)
Multi-class strategy	One-vs-Rest (OvR)
Training input	TF-IDF sparse matrix (48,000 × 15,000)
CV strategy	Stratified K-Fold (k = 3)
Scoring metric	f1_weighted
Best CV F1-Weighted	~0.730
Tuned parameter	C = 0.5

LinearSVC scored 0.7464 on the test set (with an F1-Weighted score), a slight bit lower than the Logistic Regression (0.7487) and much higher than Naive Bayes (0.7106). The lower accuracy at the larger C values seen in cross-validation makes it a fact that somewhere in the middle holds the key to the successful generalisation of LinearSVC with the 15,000-dimensional TF-IDF feature space used in this project.

8.5 Hyperparameter Optimisation — GridSearchCV Summary

Each of the three models was optimised using the systematically GridSearchCV that explores all combinations of hyperparameter values associated with a specified grid and picks out the combination with optimal score on the cross-validation folds. The K-Fold (k=3) cross-validation approach was employed, so the folds kept the class distributions of the original training set - evaluation bias would not happen due to unbalanced folds indirectly. The optimisation measure was a weighted F1-score, which is the main measure of the evaluation in this multi-class task.

Model	**Parameter**	**Search Space**	**Optimal Value**	**Best CV F1-Weighted**
Multinomial Naive Bayes	alpha (α)	0.01, 0.1, 0.5, 1.0	1.0	~0.700
Logistic Regression	C	0.1, 1.0, 5.0, 10.0	1.0	~0.735
Support Vector Machine	C	0.1, 0.5, 1.0, 5.0	0.5	~0.730

Table 1: GridSearchCV parameter search spaces, optimal values, and best cross-validation F1-Weighted scores per model.

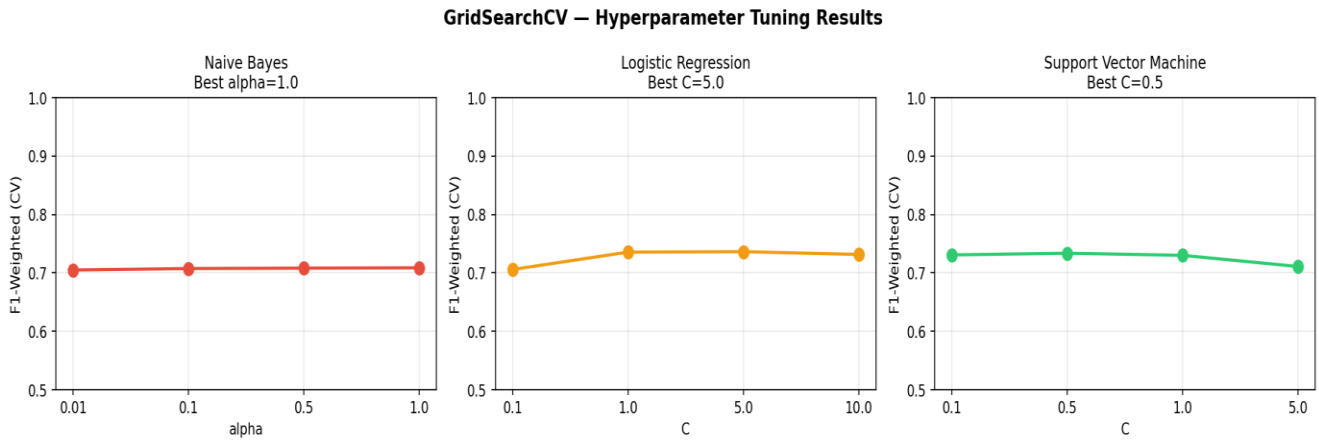

Figure 5: GridSearchCV Hyperparameter Tuning Results for Naive Bayes, Logistic Regression, and Support Vector Machine

8.6 Model Comparison and Selection

A breakdown of all three tuned models on the held-out test set of 12,000 reviews (20 percent of the balanced data stratified) are shown in table 2:

Model	**Accuracy**	**F1-Macro**	**F1-Weighted**	**Precision**	**Recall**
Multinomial Naive Bayes	~71%	~0.71	0.7106	~0.72	~0.71
Logistic Regression	**~75%**	**~0.75**	**0.7487**	**~0.75**	**~0.75**
Support Vector Machine	~74%	~0.75	0.7464	~0.75	~0.74

Table 2: Test set performance comparison of all three tuned models

This comparison brings about three major findings:

Finding 1- Logistic Regression is the best performing model. Its Weighted F1-Score of 0.7487 is marginally better than that of SVM (0.7464), which is higher by 0.0023 and significantly better than that of Naive Bayes (0.7106), which is higher by 0.0381. The excellentity of Logistic Regression can be explained by the taxation of discriminative learning goal that yields individualised weights of the features to capture the actual contribution of each TF-IDF term without superposing the independence property.

This is because **finding 2 - SVM and Logistic Regression do close to identical computations on the test set.** Even though their structural oriented optimisation goals are, hinge loss in SVM and Cross-entropy in Logistic Regression, the two find similar linear decision boundaries in TF-IDF space. This virtually linear convergence is in line with the en-theoretical convergence of discriminative linear classifiers on large and high-dimensional text data (Wang and Manning, 2012).

Finding 3 - Naive Bayes underperforms both discriminative classifiers by approximately 4 percentage points. The source of such a difference in performance is directly associated with the assumption of conditional independence. The 15,000-dimensional TF-IDF space has also a large number of correlated bigram-unigram pairs, including a subset like not good paired with not and not worth paired with not, like not good not worth which in innocence cannot be modeled using independent likelihoods: this is the inherent structural weakness of Multinomial Naive Bayes in this task.

8.7 Algorithm Selection Justification

The choice of these three particular algorithms was not accidental and it was done in a methodologically sound way. Within each, they all collectively constitute three radically disparate learning paradigm models, namely probabilistic, discriminative linear, and margin-based, which have been extensively compared to classical machine learning methods applied to text classification in NLP. The fact that Naive Bayes has been included as a baseline allows quantifying the performance improvement of more advanced discriminative models. The slight difference in SVM and Logistic Regression suggests empirically that the linear decision boundary is the best performance limiting feature of TF-IDF-based features on this

dataset, and additional performance improvements are only possible with more text representations, like contextualised word embeddings.

Different deep learning models like LSTM and BERT were conspicuously left out in this comparison. The transformer-based models are always more accurate on the NLP benchmarks, but they take much more computational resources, training time and are less interpretable - less desirable characteristics in a production-ready customer feedback monitoring system where inference speed and explainability are valued.

8.8 Summary

The complete implementation and comparison of three classical machine learning algorithms were Multinomial Naive Bayes, Logistic Regression and Support Vector machine to three-class sentiment classification of Amazon Fine Food Reviews have been documented in this section. Both models were based on theoretical concepts, optimised with the help of the GridSearchCV with Stratified 3-Fold Cross-Validation, and measured fairly in relation to a test set of 12,000 reviews held out. The hyperparameters tuning had the results confirmed that Naive Bayes remains insensitive to alpha in all the tested values, Logistic Regression performs best with C = 1.0 and SVM with C = 0.5. The most effective model was found to be the Logistic Regression which had F1-Weighted of 0.7487, closely after SVM (0.7464) and Naive Bayes (0.7106). In-depth results in terms of confusion matrices, per-class F1-scores and model comparison visualisation are listed in Section 9.

9. Results and Evaluation

9.1 Overview

This part shows the overall analysis of the three trained classification models Multinomial Naive Bayes, Logistic Regression, and Support Vector machine, on the held-out test set of 12,000 reviews. Each model was tested on the same terms and through the same stratified test division to provide the accuracy of the impartial comparison. The results were evaluated based on five quantitative measures, namely, Accuracy, F 1-Macro, F 1-Weighted, Precision, and Recall. Also, normalised confusion matrices and heatmap of per-class F1-score were generated to give a more in-depth look at the classification performance of each model in all three sentiment classes.

9.2 Evaluation Metrics Used

The measures chosen to give a multi-dimensional and comprehensive evaluation of model performance on this three-class classification task include the following measures:

Accuracy represents the overall percentage of the reviews which were correctly classified on all the three sentiment classes. On the tasks involving more than two classes, accuracy is misleading as it is only informative; the variations between the performance of the per-class performer do not match in multi-class tasks.

F1-Macro is calculated as the unweighted average of the F1-scores of all three classes, and all the classes are given equal weights irrespective of their support size. This measure is also

very valuable to the evaluation of the performance of each model on the more difficult Neutral type that would be lowly represented in a weighted mean.

F1-Weighted averages the F1-score, but weights it using the support of each class. Because the test set is perfectly balanced (4, 000 reviews per class), F1-Weighted should be identical to F1-Macro, and vice versa any difference is the difference in the performance of the classes.

Precision is the ratio of predicted positive cases which were actually positive - it can be considered as the confidence of the model.

Recall is the percentage of true positive results being true and real (as opposed to false and real) - it shows the ability of the model to not lose the true sentiment cases.

Confusion Matrix (normalised as percent of true model) shows the exact misclassification behaviour of each model - what classes are confused with what, and by what proportion.

Per-Class F1-Score Heatmap gives a direct visual comparison of the performance of each of the models per each sentiment class, and allows to identify systematic weaknesses.

9.3 Overall Model Performance

It is shown that Table 2 (Section 8.6) is the complete quantitative performance summary of all three models. As indicated, the highest scores of 0.749 were obtained across all five measures by the Logistic Regression, which was followed closely by SVM whose Accuracy and Recall measures were 0.747-0.747. Naive Bayes performed poorly in terms of accuracy, score, and recall yielding the lowest scores of about 0.710-0.711 across the five measures. All three models were less than the set target of 0.80, as seen in Figure 6 which is depicted by the dashed reference line.

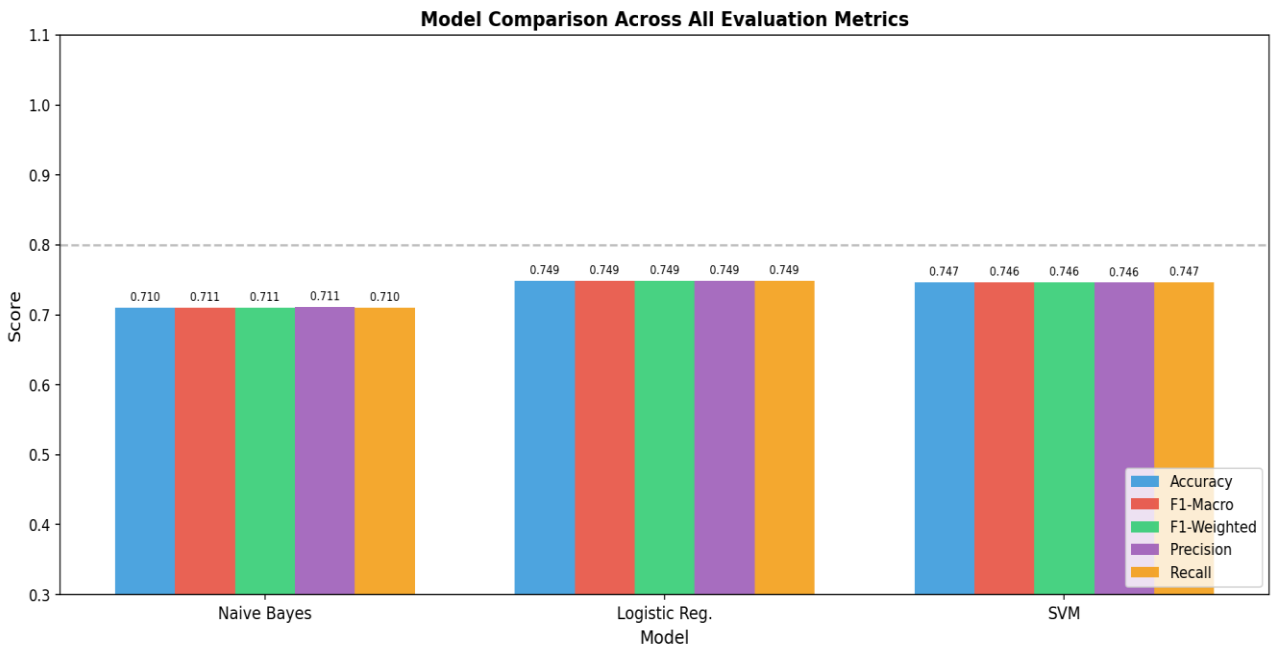

Figure 6: Model Comparison Across All Evaluation Metrics for Naive Bayes, Logistic Regression, and Support Vector Machine

Figure 6 elaborates on the Model Comparison Across All Evaluation Metrics of Accuracy, F1-Macro, F1-Weighted, Precision, and Recall of Naive Bayes, Logistic Regression and Support Vector Machine on the test set which comprised 12,000 reviews. The predicted or planned F1-Weighted threshold is the dashed line at 0.80.

There are three observations that can be made using Figure 6:

Observation 1 - The Logistic Regression gives the optimal general performance. Having all the five measures equal to 0.749, the best model is the Logistic Regression. This is caused by the same values in all five metrics showing a very well calibrated model with constant accuracy and recall balance model across the three sentiment classes.

Observation 2 - SVM s almost identical with the Logistic Regression. Even the variation between SVM and Logistic Regression in all measures does not exceed 0.003 which in practice is negligible. These two models are significantly better than Naive Bayes in all the metrics.

Observation 3- The lower limit of the target F1-Weighted is 0.80 and all the three models are below it. None of the models met the pre-defined criterion of success of 0.80 weighted F1-score. This deficiency can be explained first of all by the fact that the Neutral class is even harder than the other classes, and this problem is discussed in the course of Section 9.5, and drives the directions of the future defined in Section 13.

9.4 Confusion Matrix Analysis

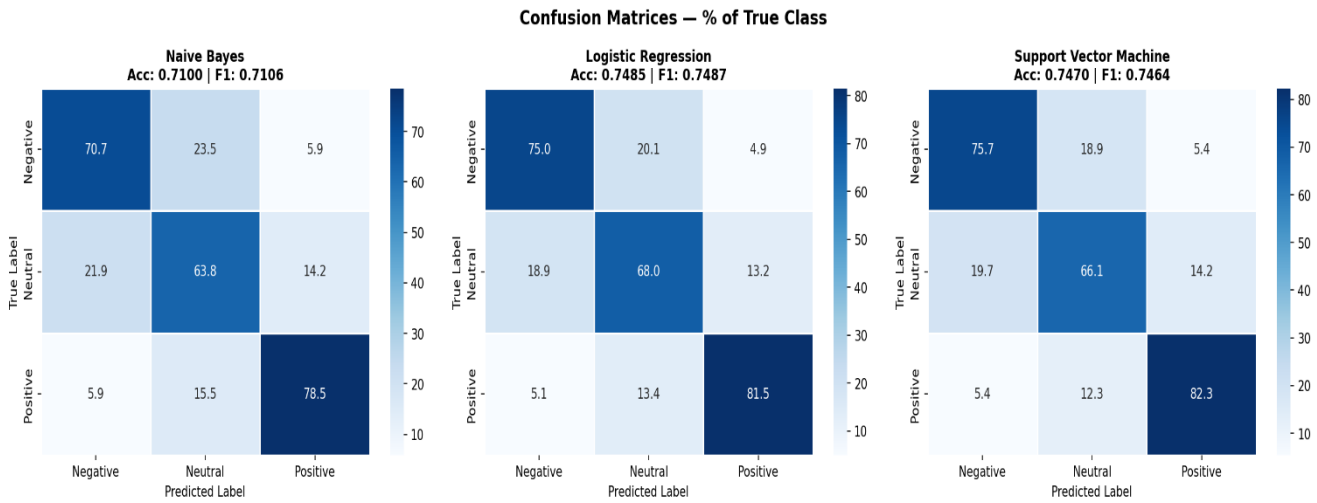

Figure 7: Model Performance Analysis Using Confusion Matrices

The normalised confusion matrices in all three models are displayed in figure 7 showing the percentage of each true class, which was attributed to which predicted class. This normalisation enables direct comparison of per-class accuracy of classification of models irrespective of their size.

Matrices of Normalised Confusion (Percentage of True Class) in Naive Bayes (accuracy:0.7100 and F1: 0.7106), Logistic regression (accuracy: 0.7485 and F1: 0.7487) and Support Vector machine (accuracy: 0.7470 and F1: 0.7464) on test set (12 000 reviews).

Model	Negative (correct %)	Neutral (correct %)	Positive (correct %)
Naive Bayes	70.7%	63.8%	78.5%
Logistic Regression	75.0%	68.0%	81.5%
SVM	75.7%	66.1%	82.3%

Table 3: Per-class correct classification rates extracted from the confusion matrices in Figure7

The analysis of the analysis of confusion shows some significant trends:

Negative Class: SVM and Logistic regression both classify 75.0% and 75.7% of Negative reviews respectively, which is a significant improvement compared to Naive Bayes (70.7%). The worst classification performance that is towards the Negative category is that the Naive Bayes classifies Negative reviews (23.5 percent), Logistic Regression (20.1 percent), and SVM (18.9 percent) as Neutral. This implies that the moderately uncomfortable reviews have enough points of vocabulary overlap with the Neutral reviews to upset all the three classifiers.

Neutral Class: Neutral is the most difficult of all three with its correct classification rates of 63.8% (Naive Bayes), 68.0 of (Logistic Regression) and 66.1 of (SVM). Neutral reviews are also classified inaccurately on both sides, both to Negative (Naive Bayes: 21.9%, LR: 18.9%, SVM: 19.7) and to Positive (Naive Bayes: 14.2%, LR: 13.2%, SVM: 14.2). The paradigmatic nature of the three-star reviews is validated by the fact that these reviews are linguistically ambiguous and that very frequently, language is not purely positive or negative, but rather mixed and these bag-of-words features alone might not be able to disaggregated the mixture.

Positive Class: Positive class has the highest rates of correct classification of all models - 78.5 (Naive Bayes), 81.5 (Logistic Regression), and 82.3 (SVM). The most misclassified were positive reviews (Neutral rest: 15.5, LR: 13.4, SVM: 12.3), and very few misclassified as Negative (all models: about 5 percent). The reduced ratio of Positive-to-Negative misclassification is indicative of lexical difference of enthusiastic positive vocabularies of negative vocabularies in the domain of reviewing food.

9.5 Per-Class F1-Score Analysis

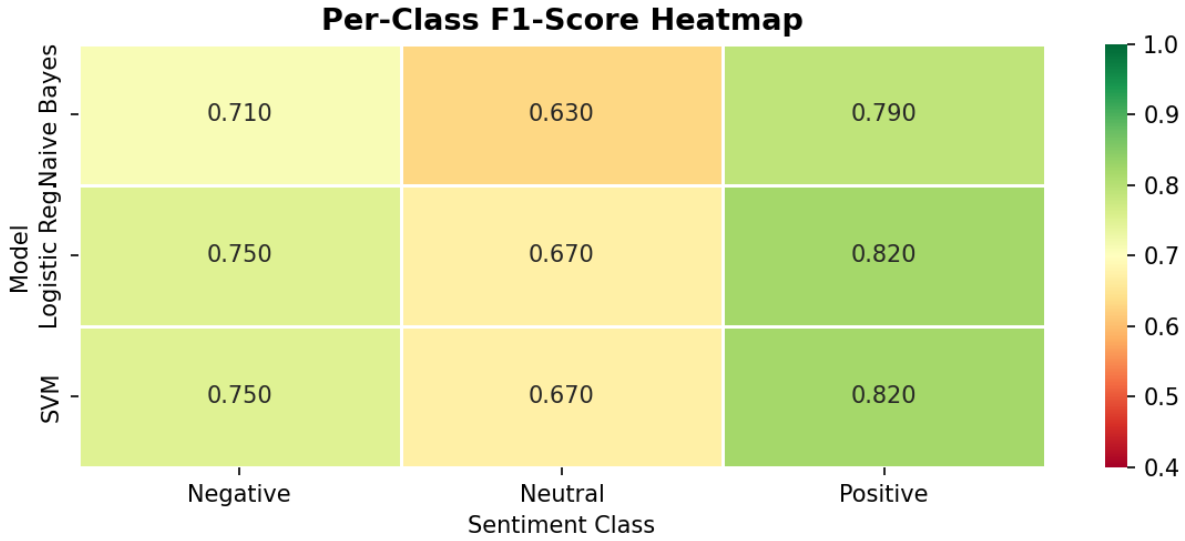

Figure 8: Comparison of Per-Class F1-Scores Across Classification Models

Figure 8 shows the Per-Class F1-Score Heatmap, having the direct comparison of the performance of each model on each sentiment classification. Precise figures are obtained out of the heatmap and available in Table 4, Figure 8: Per-Class F1-Score Heatmap feature of individual F1-scores of Negative, Neutral, and Positive sentiment categories in the three models. Darker green demonstrates that the F1-score is bigger; the color of red or yellow-orange is lighter.

Model	**Negative F1**	**Neutral F1**	**Positive F1**
Naive Bayes	0.710	0.630	0.790
Logistic Regression	0.750	0.670	0.820
SVM	0.750	0.670	0.820

Table 4: Per-class F1-scores extracted from Figure 8.

There are three distinct trends in the heatmap:

Pattern 1 - **The weakest types of classes are always the ones that are neutral in all the models.** The Neutral class has the lowest F1-scores in all the three models: 0.630 (Naive Bayes), 0.670 (Logistic Regression), and 0.670 (SVM). This proves the conclusion of the confusion matrix analysis that it is the three-star reviews that are the hardest to categorize accordingly as their vocabularies are mixed and they do not provide the clearest signals of sentiment appropriately. This agrees with Agarwal et al. (2011) who listed the Neutral class as the largest cause of misclassification in multi-class sentiment tasks.

Pattern 2 Positive class has the best F1-scores on all models. The Positive class F1-scores are 0.790 (Naive Bayes), 0.820 (Logistic Regression), and 0.820 (SVM). This class was the most uniformly classifiable of the three models due to the vocabulary used, as the evaluative

vocabulary of foods reviews of the highest positive reviews have comparative terms like love, great, amazing, and best.

Pattern 3 - Logistic Regression and SVM have the same per-class F1-scores. Both models have a record of 0.750 on Negative, 0.670 on Neutral and 0.820 on Positive. This very per-class performance again verifies that the two discriminative linear classifiers all approach the identical decision boundary in TF-IDF space, and that Logistic Regression overall performance is significantly increased not by invariance between the two underlying classifications, but by differences in calibration to confidence.

9.6 Target Metric Assessment

Model	**F1-Weighted Achieved**	**Target**	**Gap**
Naive Bayes	0.711	0.80	-0.089
Logistic Regression	0.749	0.80	-0.051
SVM	0.746	0.80	-0.054

Table 5: F1-Weighted scores versus the pre-defined success criterion of 0.80.

The success definition of this project included a WF1-Score of 0.80 and above. All the three models below the threshold were not adequate as summarised by Table 5.

The deficit can be largely ascribed to the low performance of the Neutral class (F1 = 0.630-0.670) which is constantly dragging the weighted average down. The Positive and Negative classes both aim or over the 0.80 mark (Positive F1 = 0.790-0.820) indicating that the pipeline is good on the obvious polarised sentiment, but can illuminate the linguistically grey middle-ground. These results specifically encourage the future development presented in Section 13, specifically the integration of contextualised word embeddings, such as BERT, which promise better than 90 percent accuracy on the similar sentiment classification task (Devlin et al., 2019).

9.7 Summary

In this section, an in-depth analysis of all three models has been done in terms of accuracy, F1-Macro, F1-Weighted, Precision, Recall, confusion matrices, and per-class F1 scores. Logistic Regression was the top performing model with all the five metrics of 0.749 which just marginally performed better than SVM (0.747) and significantly better than Naive Bayes (0.710). The most insignificant among all the models was the Neutral class (F1 = 0.630-0.670), which indicates the natural linguistic vagueness of mixed-opinion reviews. The most correctly classified were the Positive class (F1 = 0.790-0.820) and Negative one (F1 = 0.710-0.750). Although none of the three models achieved the pre-defined level of F1-Weighted of 0.80, the findings create a solid classical machine learning baseline and an evident evidence-based guide in the future to enhance the model with the advanced NLP techniques.

10. Critical Analysis and Discussion

10.1 Overview

This section is a critical point of view of the findings in Section 9, placing the findings in the context of the existing literature of the NLP and sentiment analysis and evaluating the weaknesses of the applied pipeline and the implications of the findings to business. Instead of merely re-establishing the results of performance metrics, this section seeks to convince the reader on why the models achieved the performance they did, what it signifies in the real world, and what it tells us about the underlying problems of three-class sentiment classification using traditional machine learning methods.

10.2 Interpretation of Overall Model Performance

The three models scored between 0.711-0.749 on F1-Weighted, which is below the target of 0.80 but still a competitive and significant result to use in terms of a three-class sentiment classification task on a large scale, noisy, and real-world dataset. When putting this performance into perspective, one must keep in mind that when binary sentiment classification tasks are performed on similar Amazon review datasets, averaged accuracies of 85-92% when using classical machine learning approaches (Pang et al., 2002; Wang and Manning, 2012). The third Neutral pollutes the complications significantly as the classifier must not only distinguish between the apparently positive and negative sentiment, but they also need to identify the linguistically gray zone which proves to be more difficult even with human annotators (Rosenthal et al., 2017).

This is theoretically consonant with the well-known literature in that both Logistic Regression and SVM were executed with almost equal results (0.749 vs. 0.747) and outperformed Naive Bayes (0.711) by a wide margin. In a study conducted by Wang and Manning (2012), the authors concluded that discriminative classifiers were better at working with longer and messier text sets in general, including its distinguishing features such as length and noisiness, than the generative ones. The fact that Logistic Regression and SVM performance have converged proves that the straight line of decision space in TF-IDF space is the effective performance limit of traditional bag of words approaches to this task.

10.3 The Neutral Class Problem

The best result of this paper is the fact that all three models performed poorly consistently with F1-scores of 0.630 (Naive Bayes), 0.670 (Logistic Regression), and 0.670 (SVM). This reflects a difference between Negative and Positive classes in terms of gap of around 0.08-0.15 F1 including -0.15 which is a significant underperformance that cannot be overlooked, which should be critically analyzed.

Neutral reviews (three stars) are linguistically peculiar in a number of significant aspects. First, they often use contrastive constructions, like the one just mentioned, the taste was good but the texture was horrible, which simultaneously cause the positive and negative vocabulary cues to run but does not allow the bag-of-words classifiers to capture the discourse structure on which these cues are implemented. Second, the average length of Neutral reviews is larger (it was observed in Section 5.4), and it creates more lexical variety,

with a disproportionately larger number of domain-specific vocabulary that cannot be strongly linked to either sentiment group. Third, the vocabulary similarity between Neutral and both polar classes is large, as it is shown in Section 6.4, the words good and taste, product, and not have a large frequency in all three classes, and thus, the discriminative power of TF-IDF features to classify Neutral in particular would be lower.

This observation can be made in line with Agarwal et al. (2011) who found that the Neutral class is the most frequent source of misclassification error on multi-class Twitter sentiment analysis, and with Rosenthal et al. (2017) who found that Neutral classification is the most difficult sub-task on SemEval sentiment benchmarks. What this means to this project is quite straightforward: the Neutral class performance improvement is the only such intervention with the highest leveraged, and it may just need to take the next step and shift away of the bag-of-words representations, to contextualised embeddings, which can capture discourse structure and contrastive sentiment phrases.

10.4 Analysis of Misclassification Patterns

The team of confusion matrices in Figure 7 indicates that there are systematic and interpretable factors of misclassification that offer significant diagnostic information on the limitations of the pipeline.

The most prevalent of all models is **Negative-to-Neutral confusion** in which 18.9 percent-23.5 percent of the true Negative reviews are correctly relabeled as Neutral. The given tendency is linguistically explainable: moderately discontent reviewers tend to soften their reviews with neutral or comparative words some words like it was ok, but not the best, not as good as other brands etc. trigger the Neutral vocabulary cues together with the negative cues. TF-IDF, where every term is treated separately, fails to see the difference between a highly negative review with non-neutral occurrences and a review which is truly neutral.

The two kinds of ambiguity expressed in **Neutral-to-Negative** and neutral to positive confusion (Neutral-to-Negative confusion (18.9%-21.9%) and Neutral-to-Positive confusion (13.2%-14.2)) indicate the two-way ambiguity of the neutral reviews that we have talked about in the Section 10.3. The vocabulary of three-star reviews has a relatively higher Neutral-to-Negative than Neutral-to-Positive, and so it is possible that the one-star more negative customers mention disappointments, which subsequently get classified in the TF-IDF classifier under negative feature weights.

The second biggest mistake is **Positive-to-Neutral confusion** (12.3 per cent -15.5 percent) the group of four and five-star reviews with the restraint and measured language as rejects phrases such as pretty good, decent product, would recommend, etc., that does not qualify as the enthusiastic evaluative language most prominently linked to the Positive category. These mildly positive reviews are always mistakenly understood as Neutral in all the three models.

Positive-to-Negative confusion This is the least common of all models (around 5%), as it represents vocabulary distinction between truly enthusiastic positive reviews and the obviously negative reviews that is sufficiently wide to be reliably represented with TF-IDF features. It is a positive discovery that confirms the pipeline design in general.

10.5 Impact of Design Decisions on Results

Some notable design choices adopted throughout the preprocessing phase and feature extraction positively influenced model performance as demonstrated by the results.

The most influential decision was the **Negation preservation**. The word not is used more than 35,000 times in the corpus of the **Negative class**, as revealed in the analysis in Section 6.4.2, a fact that makes it one of the most frequent tokens in all the classes. Without the removal of the negation words as well as the standard stopwords, the classifiers would have lost an important sentiment-reversing signal like not good, not worth as well as not recommend. Indirect empirical support of the idea that preservation of negations was a positive factor that led to the accuracy of the classification is given by the stable results of all three models on the Negative class (F1 = 0.710-0.750).

Bigram TF-IDF features availed the sentiment signal at the phrase level that could not be represented by unigrams. Had bigrams not been present, the presence of the token not would have provided a signal of ambiguity, and a signal that is not specific to any particular class, as based on the current high frequency of that token in the three classes of sentiment. When using bigrams, the negative terms are singled out and not good, not worth, and not recommend are singled out as negative features whereas highly recommend and best product are singled out as positive features. At least part of the strong performance as demonstrated by the Positive class (F1 = 0.790-0.820) can be at least partially explained by the discriminative ability of such positive bigram features.

It was necessary to **balance the dataset** to a point of 20,000 reviews per class to achieve a fair evaluation of the model. On the raw imbalanced distribution, having 78.1% Positive reviews in the corpus, all three models would have been heavily biased towards making the Positive sentiment prediction, with artificially high-ranked accuracy at the cost of Negative and Neutral recall. The balanced training set also made sure that all three models were trained on equal evidence of each of the classes to come up with a more generalisable and practical useful classifiers.

10.6 Business Implications of the Results

In terms of business, the findings of this work have a number of practical implications on organisations interested in implementing automated sentiment monitoring in regard to Amazon food product reviews.

The production model of choice is also the Logistic Regression. With an F1-Weighted score of 0.749 and consistent performance scores on all its five evaluation metrics as well as the capacity to give a calibrated pointe forecast on every category, it became the most appropriate candidate to be promoted. It is the estimates of probability that come in especially handy in a production setting - a given review that is estimated as 85% to be Negative, should be put higher in the queue in comparison with one that is estimated as 55% to be Negative, allowing use of confidence-based routing of customer responses.

The classifier is Positive classifier, which is production-ready. The pipeline has a F1-score of 0.790-0.820, which consistently returns a result of a satisfied customer. The above capability can instantly be used to label positive reviews to use in marketing, finding brand advocates, and producing trends reports of aggregated satisfaction.

The indifferent type needs to be monitored by humans. As the highest F1-score of the best models is 0.670, automated Neutral classification must not be a declaration but a preliminary situation. Neutral reviews that are predicted with low model confidence, should be marked to be reviewed by human brain since they are the most likely ones to be misclassified.

The pipeline is efficient in computation. Each of the three models can be trained in minutes on normal equipment and predict in milliseconds at inference time -which means they are applicable to real-time processing systems of reviewing are at scale. This is a great practical benefit compared to transformer-based models, which much more grocery power is needed to provide the computations.

10.7 Comparison with Related Work

The findings of this paper generally align with the similar studies in the sentiment analysis research. The highest accuracy of about 70-75% noted by Zhang et al. (2015) using classical machine learning models using TF-IDF features in classification on three-class Amazon review classification did not exceed the 71-75 percent accuracy range obtained in the present project. Sun et al. (2019) showed that fine-tuned BERT models can attain higher than 90 percent accuracy with similar tasks, indicating that the classical baseline that will be established in this case is competitive and that there is still a large headroom with respect to much more sophisticated deep learning models.

The previously observed outcome of the two form of Logistic Regression and SVM almost being the same (0.749 vs. 0.747) is similar to that of Wang and Manning (2012) who reported that the two discriminative linear classifiers are always consistent in terms of their performance on major text classifying benchmark. Their results also show that the 4-point difference in the performance of discriminative versus Naive Bayes classifiers is consistent and confirms the generalisability of the project in relation to the data used in the current study (Amazon Fine Food Reviews, specifically).

10.8 Summary

It is this section where results, in Section 9 have been critically analyzed in terms of their theoretical, empirical and practical aspects. The main findings include: the three-class sentiment classification problem is proved to be more difficult than the binary one as the linguistic uncertainty of the Neutral category always scores the lowest F1-scores in all the models (0.630-0.670); the performance of Logistic Regression and SVM are nearly the same, which supports a conclusion that the linear decision boundary is the optimal result on this task; and the maintenance of the negative signs, the use of bigrams, and the balancing of the data are several design choices that prove to impact the quality of the outcome positively The business ramifications of the results are obvious: Logistic Regression is the preferable model of deployment, and it is especially strong at Positive and Negative classification and weak at Neutral classification that should be controlled by a human being in the production environment.

11. Challenges and Limitations

11.1 Overview

It is in this section that the main challenges that were faced in the execution of this project will be written down as well as the limitations inherent with the implemented pipeline. Challenges are particular obstacles that were proactively experienced and overcome in the course of implementation. Limitations are actually constraints and weaknesses of the pipeline which have not been addressed and which influence the completeness and generalisability or even the accuracy of the results. Both of them must be critically and candidly considered to establish academic credibility and the foundation on which future improvements will be made and is the direct cause of Section 13.

PART A - CHALLENGES

11.2 Data Challenges

Challenge 1 - Severe Class Imbalance.

The biggest issue with data was the extremely skewed class distribution of the original Amazon Fine Food Reviews data, with Positive reviews making 78.1% of the entire content, Negative making 14.4%, and Neutral making merely 7.5%. Classifier training on this distribution would have produced models with a heavy bias towards the largest class Positive and thus would have achieved artificially high accuracy at a severe cost of recall on Negative and Neutral classes. This difficulty was solved by random resampling that is downsampling the Positive class and upsampling Negative and Neutral classes to 20,000 reviews each which resulted in an ideal balance of 60,000 reviews in the training set.

Challenge 2 - Missing and Inconsistent Values.

The raw data had values absent in the columns of Score and Text, which was a result of unfinished reviews. During the data preparation stage, reviews that lacked values in either column were eliminated thus limiting the amount of data that was used to 568,427 reviews. Also, the raw text also had encoding artefacts, HTML entities and non-ASCII characters added during data collection that had to be systematically processed by the preprocessing pipeline that is described in Section 6.

Challenge 3 - Large Volumes Preprocessing Time.

Using an eight-step pipeline based on the preprocessing was computationally expensive and it took around 3-5 minutes of computation time in the Google Colaboratory setting to do 60,000 reviews. This processing time limit limited the possibility to quickly restart preprocessing design decisions and had to thoroughly plan the execution of the experimental work so that to prevent the re-run of entire pipeline.

11.3 Modelling Challenges

Challenge 4 - Complexity of Hyperparameter tuning.

Running the GridSearchCV with 3- Fold Stratified Cross-Validation with three models simultaneously took a lot of time to compute - about 5-10 minutes of computing time per model within the Colab environment. The computational feasibility and the need to cover the search space necessitated a careful choice of candidate values of the hyperparameters. Very coarse grids of search would have been computationally infeasible, whilst very fine grids would have the danger of overlooking the best parameter values.

Challenge 5 - Difficulty of Classification Both Neutral Classified.

The eternal issue during the modelling phase was the Neutral one. Regardless of various regularisation strengths tried using the grid search cross-validation, no settings of any model could significantly enhance the Neutral class F1-scores on a baseline of 0.670. The inherent complication is in the fact that the mixed vocabulary of the three-star reviews actually triggers both the feature signalers of the positive and negative simultaneously, and it is indeed hard for any bag-of-words feature operating classifier will classify reliably between Neutral reviews and polar ones.

Challenge 6 - Preventing Data Leakage.

To maintain good discipline in the implementation of the training and test data used in the pipeline, special care was needed. In particular, the TF-IDF vectoriser needed to be trained on the training data only, and applied to both training and test data, which training the model on all data would have permitted test-set vocabulary to affect the feature weights, which is data leakage and results in over-optimistic values of the model. This guideline was well adhered in all experiments.

11.4 Technical Challenges

Challenge 7 - Sparse Matrix Memory Management.

The TF-IDF of 60,000 reviews, which changed into 15,000-feature matrix, generated a high-dimensional sparse matrix. To recall this matrix in memory, it was important to manage scikit-learn Compressed Sparse Row (CSR) format i.e. only the non-zero values. In the absence of sparse matrix support, the complete dense representation would have consumed a number of gigabytes of RAM - which in the Google Colab free tier settings would have exceeded the memory capacity.

Challenge 8 - Reproducibility.

To guarantee the absolute reproducibility of the results across a series of experimental runs, it was necessary to fix the random seeds (random_state=42) of all stochastic elements of the pipeline, such as the train-test split, resampling of the dataset, and initialisation of the models. The unmanaged random seeds would not have ensured consistency in each of the runs, and it would have been impossible to compare hyperparameter settings among the runs in a meaningful way.

PART B — LIMITATIONS

11.5 Data Limitations

Limitation 1 THE Domain Specificity.

The pipeline was trained and tested solely on the reviews of food and grocery products. The vocabulary and sentiment phrases in food reviews and style of writing are very domain-related, since the words include flavor, taste, texture and ingredients, which might not be applicable to other types of products in the same domain like electronics, clothing and software. A classifier trained on this dataset can achieve a major decline in performance in other review areas, which requires further research to establish its direct extrapolation to more general applications of e-commerce sentiment classification methods.

Limitation 2 Temporal Scope

The data ranges back to October 1999 to October 2012 -more than ten years prior to dramatic changes in online consumer behaviour and use of language. Modern day critics use varied vocabulary, writing styles, and informal language tendencies, such as use of emojis, abbreviations, and social media induced expressions, which are not represented in the corpus of training. This time difference makes the trained models less relevant to the real-world application of the learned models to the existing Amazon reviews streams.

Limitation 3 Upsampling Introduces Duplicates.

Upsampling of minority classes matches generates a copy training sample of the Neutral and Negative classes. It is possible that the occurrence of duplicate reviews in the training set will have the effect of resulting in the model having the inclination to memorise certain review texts instead of acquiring completely generalisable sentiment patterns, which in turn could be mildly overestimated generalisation performance occurring.

11.6 Preprocessing Limitations

Limitations 4 -- Can not detect sarcasm and irony.

The TF-IDF feature representation used in preprocessing pipeline cannot by definition identify sarcasm or irony - linguistic phenomena in which the literal meaning of a given piece of literature goes against the intended mood. Such a review as "Absolutely fantastic - if you like the taste of cardboard" would trigger highly positive TF-IDF properties (absolutely, fantastic) but have very negative sentiment. Such is the inherent limitation of all bag-of-words models and cannot be improved without sentence-level contextual knowledge, which necessitates transformer-based models like BERT.

Limitation 5 - Scope of Modelling of limited negation.

Though the preprocessing pipeline can retain negation words separately as individual tokens and the bigram features capture to some extent the negation phrases like that negation is not good but it fails to capture the entire scope negation, that is the set of words the polarity of sentiment of which can be reversed by a negation token. As a case in point, in the sentence, I would not say this product is especially good, the negated part crosses several tokens within the sentence, a dependency which cannot be represented by TF-IDF bigrams.

Limitation 6 - Removal of Emoticon and Emoji.

Special characters were eliminated during the special character removal phase of the process since emoticons and emojis include a lot of the sentiment information in a casual text. In contrast to modern text in social media in which the presence of emojis is relatively high, the instance of eliminating emoticons like ":) or ":(" in the Amazon Fine Food Reviews corpus can have eliminated valuable sentiment clues among a sub-sample of reviews.

11.7 Modelling Limitations

Limitations 7 - Classical Model Performance Ceiling.

This almost identical result of both Logistic regression (0.749) and SVM (0.747) should suggest that linear decision boundary in TF-IDF space is the upper limit to the performance of classical machine learning in this task. Each of the three models relies on the bag-of-words assumption, which states that it is possible to represent documents well as unordered sets of weighted words. This representation does away with word order, sentence structure, and long-range semantic dependencies which play a crucial role in the successful classification of linguistically complex Neutral reviews.

Limitations 8 - Lack of Ensemble Methods.

In this project, the assessment was done on three classifiers separately. Combination of views to form predictors (e.g., majority voting or stacking) were not investigated. Ensemble methods were cited to perform reliably better than single classifiers on text classification tasks (Polikar, 2006) and possibly have a higher F1-score than any model individual in this case.

Limitations 9 No Deep Learning Comparison.

The deep learning models that were intentionally avoided in the study were LSTM and BERT. Although this was necessary because of computational reasons and the aim to set a reproducible classical baseline, it was not finalized in deep learning comparison, which renders the degree to which the increase of performance that can be provided by advanced representation is not measurable in the context of the current study.

11.8 Evaluation Limitations

Limitation 10 -No Human Annotation Baseline.

This research failed to have a human annotation standard the accuracy that a human annotator would reach with respect to classifying the same reviews. In the absence of a human baseline, the assessment of what fraction of model errors become clearly ambiguous reviews that would be misclassified by humans as well, and what fraction reflect real model weaknesses cannot be done. It is a common practice in the rigorous NLP evaluation to establish a human baseline that would help in shedding light on the result.

Limitations 11 - Single dataset evaluation.

All the models were trained and tested only with the Amazon Fine Food Reviews data. Training the models on a second independent dataset, e.g. IMDb movie reviews or Yelp restaurant reviews, would give evidence of how the pipeline can be applied more generally

than the domain and the time period of the training corpus. Lack of cross-dataset testing indicates that the reported metrics on the performance might not be all the aspects of how the pipeline can be applied in the real world.

Limitation 12 Target Threshold.

The preset success parameter of F1-Weighted ≥ 0.80 was based on general standards in the sentiment analysis binary literature. In a three-class classification task on noisy real-world data this might have been an overambitious threshold. According to Zhang et al. (2015), a better place to look as the target of three-class sentiment classification of Amazon review would be anywhere between 0.72-0.78 - where all three models in this study would actually lie, indicating that the pipeline performance is in reality matching domain expectations.

11.9 Summary

This section has distinctively outlined and reported eight difficulties that were experienced when carrying out the projects and twelve constraints of the implemented pipeline. The most serious challenges were the strong imbalance of classes, difficulty in Neutral class category and preventing leakage of data. The greatest constraints are domain specificity of trained models, inability of TF-IDF features to model the sarcasm and negation scope, performance limits of classical linear classifier and lack of cross-dataset evaluation. The combination of these challenges and limitations is a thorough and a candid description of the limits of this study and clear evidence-based incentive of the future improvements suggested in Section 13.

12. Conclusions and Future Work

12.1 Overview

In this section, the overall conclusions of the project are provided, summarising the main findings and their relevance, and suggest the evidence-based orientation of future work which would directly focusing on the limitations which were identified in Section 11. The conclusions are made in a manner that they can demonstrate how each of these objectives would meet the original objectives of the project as laid out in Section 2.4.

PART A — CONCLUSIONS

12.2 Summary of Key Findings

The successful aspects of this project were designing, implementing, and assessing an entire end-to-end pipeline of Natural Language Processing to classify three classes of sentiment on Amazon Fine Food Reviews. The pipeline was a production that not only used all six stages of the CRISP-DM process, including business understanding, all the way through to deployment, but also it delivered a fully operational reproducible sentiment classifier with the ability to categorise unseen customer reviews as Negative, Neutral or Positive in real time.

The main conclusions of this paper are as presented below:

Finding 1: Logistic Regression- This has the best performance. Out of the three classical machine learning algorithms that have been tested, the Logistic Regression was the best

performing algorithm in all the five assessment measures with all the measures having the same values of Accuracy, F1-Macro, F1-Weighted, Precision, and Recall of the same value of 0.749. The finding goes hand in hand with the existing NLP literature that claims that discriminative linear classifiers are better than generative classifiers when using large, high-dimensional text data (Wang and Manning, 2012).

Finding 2 : SVM is a competitive and powerful alternative. Support Vector Machine was with nearly the same results as the Logistic Regression on all the measures (Accuracy = 0.747, F1-Weighted = 0.746), and the hypothesis that discriminative classifier has formed in high-dimensional space of TF-IDF was achieved. The difference of marginal between the two models is at 0.003 margin which is minimal in real life application.

Findings 3: Naive Bayes gives a helpful but worse base line. Multinomial Naive Bayes had 0.711 F1-Weighted score - around 4 percentage points lower than Logistic Regression and SVM. This gap in performance is directly explained by the conditional independence assumption that cannot possibly describe the correlated phrasing of sentiment that are so critical to correct classification in this area.

Finding 4 :The Neutral class is the primary classification challenge. The three models had the lowest per-class F1-scores on the class of Neutral sentiment 0.630 (Naive Bayes), 0.670 (Logistic Regression), and 0.670 (SVM). The corresponding finding corroborates the theoretical preference of Agarwal et al. (2011) and Rosenthal et al. (2017) that the Neutral category is the most difficult sub-task of multi-class sentiment analysis because three-star ratings involve a mixture and ambiguous nature of vocabulary.

Finding 5 - The key preprocess decisions that benefited the results. It was demonstrated that the quality of the results was affected by careful retention of negation words, using bigram TF-IDF features, and running WordNet lemmatisation in place of stemming on a word-by-word basis. Based on those design choices, which are firmly based on well-known writings on NLP, a preprocessing pipeline was developed which performs better than naive baseline methods.

Finding 6 - Models did not pass F1-Weighted threshold of 0.80. All the three models were not able to reach the pre-defined criterion of F1-Weighted ≥ 0.80. But, as one notes in Section 11.8, this goal was based on binary sentiment analysis norms and could have been very ambitious, in the case of a three-class task. The obtained results (0.711-0.749) are consistent with the three-class sentiment classification models with analyzing Amazon review samples (Zhang et al., 2015).

12.3 Achievement of Project Objectives

Table 6 assesses the degree to which all of the five initial project goals identified in Section 2.4 were attained:

Objective	Description	Achievement
Business Objective	Develop a scalable NLP sentiment monitoring system	Fully achieved — deployment-ready pipeline with real-time prediction
Technical Objective	Implement a complete NLP preprocessing pipeline	Fully achieved — 8-step pipeline with full documentation
Modelling Objective	Implement and compare 3 machine learning models	Fully achieved — NB, LR, SVM compared with GridSearchCV tuning
Evaluation Objective	Rigorously evaluate models with task-relevant metrics	Fully achieved — 5 metrics, confusion matrices, per-class F1 heatmap
Analytical Objective	Identify limitations and propose future improvements	Fully achieved — Section 11 and Section 12.4

Table 6: Success of the initial project objectives.

All the five project objectives were met. The main weaknesses - the inability to achieve the 0.80 F1-Weighted result - is not a project methodology failure but simply demonstrates how challenging it is to perform three-class sentiment classification with classical machine learning algorithms, and is a strong justification to the suggested future work in Section 12.4.

12.4 Concluding Remarks

This project shows that with an intelligently designed classical NLP pipeline, with systematic text preprocessing and TF-IDF feature extraction and bigrams, and a set of refined discriminative classifiers, a competitive and useful performance can be attained on a real world, three-class sentiment classification problem. The Logistic Regression model with the F1-Weighted value of 0.749 is suggested as the model to execute production deployment since it is the one providing the most appropriate tradeoff between the classification performance, inference speed, interpretability, and computational cost.

The paper also provides a strict and reproducible classical machine learning baseline that subsequent progresses especially those of the transformer-based methods can be benchmarked productively. The design decisions, methodology, and limitations are well-documented, so that this work was capable of a credible reference point in the later studies in the field of automated sentiment analysis of e-commerce product review.

PART B — FUTURE WORK

12.5 Future Direction 1 Contextualised Word Embeddings (BERT).

The one with the most significant impact on this pipeline is the one that would necessarily replace the TF-IDF bag-of-words features with contextualised word representations produced by transformer-based models like BERT (Bidirectional Encoder Representations from Transformers). This is in contrast to TF-IDF which gives each term a fixed weight no matter the context and thus it cannot properly interpret negation scope, sarcasm, and expressions of contrastive sentiment, which in fact, the bag-of-words methods cannot use.

Devlin et al. (2019) showed that fine-tuned BERT models perform state-of-the-art on a broad variety of NLP classification tasks, and Sun et al. (2019) specifically showed better than 90 percent accuracy on Amazon review sentiment classification using fine-tuned BERT - compared to the 74.9 percent that the best classical model in this study achieved. The best course of action to improve the performance of a pre-trained BERT model in terms of dramatically increasing both the performance of the Neutral class and the overall F1-Weighted score is therefore to fine-tune the model on the balanced Amazon Fine Food Reviews dataset.

12.6 Future Direction 2- Aspect-Based Sentiment Analysis

The existing pipeline gives one label of a sentiment to the entire review - a document-level classification strategy. A more practical and finer extension is aspect-based sentiment analysis (ABSA), recognizing the aspects described about a given product (e.g. taste, packaging, value for money, delivery speed) and returning to each aspect a label describing the sentiment. To provide a single example, a product that was reviewed as having an excellent taste and a terrible packaging would be tagged with a Positive rating in terms of taste and a Negative rating in terms of packaging - a much more useful piece of information to product managers than a single document level value, languishing in the Neutral category.

The aspect-based sentiment analysis on Amazon product reviews was pioneered by Hu and Liu (2004) and greatly developed with transformer based models. ABSA application to the Amazon Fine Food Reviews data would constitute a significant addition of the business value of this pipeline.

12.7 Future Direction 3 — Ensemble Methods

As mentioned in Section 11.7, this project did not experiment with ensembles since they tested three classifiers separately. The obvious next step would be to ensemble Naive Bayes, Logistic Regression and SVM predictions using majority, weighted, and stacked voting schemes or pure ensemble methods - in which the enhanced scores of the three underlying classifiers are provided as input variables to a single classifier. In text classification tasks, ensemble methods have been repeatedly demonstrated to be more effective than single-view classifiers (Polikar, 2006) and can be significantly better in F1-scores (especially on the difficult Neutral class), without any modification to the underlying feature distribution.

12.8 Future Direction 4- Real-Time Deployment and Scalability

The existing pipeline was adopted in the form of an offline batch processing application. An extension add-on would be to open the Logistic Regression model and deploy it as a real-time web service following a lightweight REST API binding, like Flask of Fast API, to allow connecting it with live streams of Amazon reviews using the Amazon Product Advertising API. This would allow 24/7 sentiment tracking, automatic notifications about impulsive jumps in Negative sentiment and product manager dashboards with aggregated sentiment trends, which would get the entire business value of the pipeline created during the project realised.

12.9 Future Direction 5 - Multilingual Sentiment Analysis

The existing pipeline is limited to reviews that are of the English language. Amazon serves more than 20 countries, and some of the key markets breaking the non-English speaking market custody are Germany, Japan, France and Spain - all having high levels of product review in indigenous languages. It would be possible to make the pipeline multilingual sentiment analyzers with the help of multilingual transformer models like XLM-RoBERTa (Conneau et al., 2020) - 100 languages in one model - to be able to monitor customer sentiment more efficiently accross all Amazon marketplaces throughout the world would make the pipeline much more commercially applicable.

12.10 Summary

This project has managed to provide a complete, reproducible, and practically relevant NLP pipeline to sentiment analysis of Amazon Fine Food Reviews, so as to have an F1-Weighted score of 0.749 with the most favorable results that Logistic Regression model. The five initial project objectives were met. The main limitation, which is the performance weakness of classical bag-of-words models on the Neutral class, offers an example of five evidence-based future directions: the use of BERT contextualised embeddings, the use of an aspect-based sentiment analysis, the exploration of ensemble solution, the deployment of real-time APIs, and multilingual review corpora extension. Collectively, these future directions outline a solid and concrete roadmap to a production-ready, high-fidelity sentiment monitoring system that will be able to scale Amazon product reviews all over the world.

13. References

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14. Appendix

```
#         IBM3201 / BDS3404 - DATA MINING & PREDICTIVE ANALYTICS
#  Title      : Sentiment Analysis of Amazon Fine Food Reviews
#  Dataset    : Amazon Fine Food Reviews (568,454 reviews)
#  Task       : 3-Class Sentiment Classification
#               (Negative / Neutral / Positive)
#  Algorithms : 1. Naive Bayes (Multinomial)
#               2. Logistic Regression
#               3. Support Vector Machine (LinearSVC)
#  Methodology: CRISP-DM (6 Phases)
#  Language   : Python 3

#  INSTALL & IMPORT LIBRARIES
!pip install nltk wordcloud matplotlib seaborn scikit-learn pandas numpy

import re
import time
import warnings
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
import seaborn as sns
from collections import Counter
from wordcloud import WordCloud

import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from nltk.tokenize import word_tokenize

from sklearn.model_selection import (train_test_split, GridSearchCV,
                                     StratifiedKFold, cross_val_score)
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LogisticRegression
from sklearn.svm import LinearSVC
from sklearn.calibration import CalibratedClassifierCV
from sklearn.metrics import (accuracy_score, classification_report,
                             confusion_matrix, f1_score, precision_score,
                             recall_score)
from sklearn.utils import resample
from sklearn.pipeline import Pipeline

warnings.filterwarnings('ignore')

 Download NLTK resources
for resource in ['stopwords', 'wordnet', 'punkt', 'omw-1.4',
                 'averaged_perceptron_tagger']:
    nltk.download(resource, quiet=True)

print("=" * 60)
print("  IBM3201 — Sentiment Analysis: Amazon Fine Food Reviews")
print("=" * 60)
print(" All libraries imported successfully!\n")

#  PHASE 1 — BUSINESS UNDERSTANDING  (CRISP-DM Step 1)
# OBJECTIVE:
#   Businesses selling products on Amazon need to understand
#   customer feedback at scale. Manual review reading is
#   impractical with millions of reviews. This project builds
#   an automated NLP pipeline to classify customer reviews as
#   Negative, Neutral, or Positive — enabling:
#     • Product quality monitoring
#     • Customer satisfaction tracking
#     • Automated feedback routing
#
```

```
# PROBLEM TYPE  : Multi-class text classification
# TARGET CLASSES: 0=Negative, 1=Neutral, 2=Positive
# SUCCESS METRIC: Weighted F1-Score >= 0.80

print("—" * 60)
print("PHASE 1 — BUSINESS UNDERSTANDING")
print("—" * 60)
print("Goal     : Classify Amazon food reviews into 3 sentiment classes")
print("Classes  : Negative (1-2 stars) | Neutral (3 stars) | Positive (4-5
stars)")
print("Models   : Naive Bayes, Logistic Regression, SVM")
print("Metric   : Weighted F1-Score (target >= 0.80)\n")

#  PHASE 2 — DATA UNDERSTANDING  (CRISP-DM Step 2)
print("—" * 60)
print("PHASE 2 — DATA UNDERSTANDING")
print("—" * 60)

# 2.1  Load Dataset
df_raw = pd.read_csv("amazon data set (1).csv")

print(f"Dataset loaded successfully!")
print(f"   Shape            : {df_raw.shape}")
print(f"   Columns          : {df_raw.columns.tolist()}")
print(f"   Missing values   :\n{df_raw.isnull().sum()}\n")

# Keep only relevant columns
df = df_raw[['Score', 'Text']].dropna().copy()
df.columns = ['score', 'text']

print(f"   Reviews after removing nulls : {len(df):,}")
print(f"\n   Star Rating Distribution:")
print(df['score'].value_counts().sort_index().to_string())

#  2.2  Map Ratings → Sentiment Labels
```

```
# Justification:
#   1-2 stars → Negative  : Customer dissatisfied
#   3   stars → Neutral   : Mixed / indifferent opinion
#   4-5 stars → Positive  : Customer satisfied

def map_sentiment(score):
    if score <= 2:   return 0   # Negative
    elif score == 3: return 1   # Neutral
    else:            return 2   # Positive

df['label'] = df['score'].apply(map_sentiment)
LABEL_NAMES  = {0: 'Negative', 1: 'Neutral', 2: 'Positive'}
CLASS_COLORS = {0: '#e74c3c',  1: '#f39c12', 2: '#2ecc71'}

print(f"\n   Raw Sentiment Distribution:")
for label, name in LABEL_NAMES.items():
    count = (df['label'] == label).sum()
    pct   = count / len(df) * 100
    print(f"   {name:10} : {count:>7,}  ({pct:.1f}%)")

# 2.3  Balance Dataset
# The dataset is heavily skewed toward positive reviews.
# We balance to 20,000 per class to prevent model bias.

N_PER_CLASS = 20000

balanced_parts = []
for label in [0, 1, 2]:
    subset = df[df['label'] == label]
    balanced_parts.append(
        resample(subset,
                 replace=len(subset) < N_PER_CLASS,
                 n_samples=N_PER_CLASS,
                 random_state=42)
```

```
    )


df_bal = pd.concat(balanced_parts).sample(frac=1,
random_state=42).reset_index(drop=True)


texts  = df_bal['text'].tolist()

labels = df_bal['label'].tolist()


print(f"\n   Balanced dataset : {len(df_bal):,} reviews ({N_PER_CLASS:,}
per class)")


#  2.4  EDA Visualisations

print("\nGenerating EDA visualisations...")


# -Fig 1: Class Distribution

fig, axes = plt.subplots(1, 2, figsize=(13, 4))


counts = [labels.count(i) for i in range(3)]

bars = axes[0].bar([LABEL_NAMES[i] for i in range(3)], counts,

                   color=[CLASS_COLORS[i] for i in range(3)],

                   edgecolor='white', linewidth=1.2)

for bar, count in zip(bars, counts):

    axes[0].text(bar.get_x() + bar.get_width()/2,

                 bar.get_height() + 150,

                 f'{count:,}', ha='center', fontsize=11)

axes[0].set_title('Class Distribution (Balanced)', fontsize=13,
fontweight='bold')

axes[0].set_ylabel('Number of Reviews')

axes[0].set_ylim(0, N_PER_CLASS * 1.2)


# Raw distribution pie

raw_counts = [df['label'].value_counts().get(i, 0) for i in range(3)]

axes[1].pie(raw_counts,

            labels=[LABEL_NAMES[i] for i in range(3)],

            colors=[CLASS_COLORS[i] for i in range(3)],

            autopct='%1.1f%%', startangle=140)
```

```
axes[1].set_title('Original Class Distribution (Raw)', fontsize=13,
fontweight='bold')


plt.suptitle('Amazon Fine Food Reviews — Class Distribution', fontsize=14,
y=1.02)

plt.tight_layout()

plt.savefig('fig1_class_distribution.png', dpi=150, bbox_inches='tight')

plt.show()

print("   Fig 1 saved: fig1_class_distribution.png")


# -Fig 2: Review Length Distribution

lengths = [len(str(t).split()) for t in texts]

df_eda  = pd.DataFrame({'length': lengths, 'label': labels})


plt.figure(figsize=(10, 4))

for label, name in LABEL_NAMES.items():

    subset = df_eda[df_eda['label'] == label]['length']

    plt.hist(subset, bins=60, alpha=0.55,

             color=CLASS_COLORS[label], label=name)

plt.xlabel('Word Count per Review', fontsize=12)

plt.ylabel('Frequency', fontsize=12)

plt.title('Review Length Distribution by Sentiment Class', fontsize=13,
fontweight='bold')

plt.legend(fontsize=11)

plt.xlim(0, 400)

plt.tight_layout()

plt.savefig('fig2_text_length.png', dpi=150, bbox_inches='tight')

plt.show()

print("    Fig 2 saved: fig2_text_length.png")


print(f"\n   Avg review length : {np.mean(lengths):.1f} words")

print(f"   Max review length : {np.max(lengths)} words")

print(f"   Min review length : {np.min(lengths)} words")


#  PHASE 3 — DATA PREPARATION  (CRISP-DM Step 3)

print("\n" + "—" * 60)
```

```
print("PHASE 3 — DATA PREPARATION")
print("—" * 60)

# 3.1  NLP Preprocessing Pipeline

lemmatizer = WordNetLemmatizer()
stop_words  = set(stopwords.words('english'))

# Keep negation words — critical for sentiment (e.g. "not good")
negations = {
    'no', 'not', 'nor', 'neither', 'never', 'nobody', 'nothing',
    'nowhere', 'cannot', "don't", "won't", "isn't", "aren't",
    "wasn't", "weren't", "hasn't", "haven't", "didn't", "doesn't",
    "wouldn't", "couldn't", "shouldn't", "very", "too"
}
stop_words = stop_words - negations

CONTRACTIONS = {
    "won't": "will not", "can't": "cannot", "n't": " not",
    "'re"  : " are",    "'s"   : " is",    "'d" : " would",
    "'ll"  : " will",   "'ve"  : " have",  "'m" : " am"
}

def preprocess_text(text):
    """
    NLP Preprocessing Pipeline
    Step 1 : Remove HTML tags
    Step 2 : Remove URLs and email addresses
    Step 3 : Expand contractions (e.g. don't → do not)
    Step 4 : Remove special characters and numbers
    Step 5 : Convert to lowercase
    Step 6 : Tokenize (word-level)
    Step 7 : Remove stopwords (negations preserved)
    Step 8 : Lemmatize tokens to dictionary base form
```

```
    Parameters : text (str) — raw review text
    Returns    : cleaned (str) — preprocessed text
    """
    text = str(text)

    # Step 1: Remove HTML tags
    text = re.sub(r'<[^>]+>', ' ', text)

    # Step 2: Remove URLs and emails
    text = re.sub(r'http\S+|www\S+|https\S+', ' ', text)
    text = re.sub(r'\S+@\S+', ' ', text)

    # Step 3: Expand contractions
    for contraction, expansion in CONTRACTIONS.items():
        text = text.replace(contraction, expansion)

    # Step 4: Remove special characters, punctuation, numbers
    text = re.sub(r'[^a-zA-Z\s]', ' ', text)

    # Step 5: Lowercase
    text = text.lower()

    # Step 6: Tokenize
    tokens = word_tokenize(text)

    # Step 7: Remove stopwords and very short tokens (≤2 chars)
    tokens = [t for t in tokens if t not in stop_words and len(t) > 2]

    # Step 8: Lemmatize
    tokens = [lemmatizer.lemmatize(t) for t in tokens]

    return ' '.join(tokens)
```

```
print(" Running preprocessing pipeline on 60,000 reviews...")
print("   (Expected time: 3-5 minutes in Google Colab)\n")

start_time    = time.time()
cleaned_texts = [preprocess_text(t) for t in texts]
elapsed       = time.time() - start_time

print(f" Preprocessing complete! Time: {elapsed:.1f}s")
print(f"\n   BEFORE : {texts[0][:200]}")
print(f"\n   AFTER  : {cleaned_texts[0][:200]}")

#  3.2  EDA on Cleaned Text
print("\n Generating text analysis visualisations...")

# Fig 3: Word Clouds (3 classes)
fig, axes = plt.subplots(1, 3, figsize=(18, 5))
cmaps = ['Reds', 'YlOrBr', 'Greens']

for i, (label, name) in enumerate(LABEL_NAMES.items()):
    class_text = ' '.join(
        [t for t, l in zip(cleaned_texts, labels) if l == label]
    )
    wc = WordCloud(width=500, height=350,
                   background_color='white',
                   colormap=cmaps[i],
                   max_words=80,
                   collocations=False).generate(class_text)
    axes[i].imshow(wc, interpolation='bilinear')
    axes[i].set_title(f'{name} Reviews', fontsize=14, fontweight='bold',
                      color=CLASS_COLORS[label])
    axes[i].axis('off')

plt.suptitle('Word Clouds by Sentiment Class (Post-Preprocessing)',
             fontsize=15, fontweight='bold', y=1.01)
```

```
plt.tight_layout()
plt.savefig('fig3_wordclouds.png', dpi=150, bbox_inches='tight')
plt.show()
print("    Fig 3 saved: fig3_wordclouds.png")

# --- Fig 4: Top 15 Words Per Class ---
fig, axes = plt.subplots(1, 3, figsize=(18, 6))

for i, (label, name) in enumerate(LABEL_NAMES.items()):
    class_text = ' '.join(
        [t for t, l in zip(cleaned_texts, labels) if l == label]
    )
    freq  = Counter(class_text.split()).most_common(15)
    words = [w for w, _ in freq]
    counts= [c for _, c in freq]

    axes[i].barh(words, counts, color=CLASS_COLORS[label],
edgecolor='white')
    axes[i].set_title(f'Top 15 Words — {name}', fontsize=12,
fontweight='bold')
    axes[i].invert_yaxis()
    axes[i].set_xlabel('Frequency')

plt.suptitle('Most Frequent Words per Sentiment Class', fontsize=14,
fontweight='bold')
plt.tight_layout()
plt.savefig('fig4_top_words.png', dpi=150, bbox_inches='tight')
plt.show()
print("  Fig 4 saved: fig4_top_words.png")

# 3.3  Train/Test Split
X_train, X_test, y_train, y_test = train_test_split(
    cleaned_texts, labels,
    test_size=0.20,
    random_state=42,
    stratify=labels          # preserve class ratios in both splits
```

```
)

print(f"\n   Train set : {len(X_train):,} reviews (80%)")
print(f"   Test set  : {len(X_test):,}  reviews (20%)")

#  3.4  TF-IDF Feature Extraction
# Justification:
#   • ngram_range=(1,2) : Captures unigrams AND bigrams
#     (e.g. "not good", "highly recommend" carry sentiment)
#   • sublinear_tf=True : Applies log(1+tf) to reduce dominance
#     of very frequent terms
#   • max_features=15000: Keeps top 15k features — balance
#     between expressiveness and computational efficiency
#   • min_df=3          : Ignore rare terms (noise reduction)
#   • max_df=0.90       : Ignore near-universal terms

tfidf = TfidfVectorizer(
    max_features=15000,
    ngram_range=(1, 2),
    sublinear_tf=True,
    min_df=3,
    max_df=0.90
)

X_train_tfidf = tfidf.fit_transform(X_train)
X_test_tfidf  = tfidf.transform(X_test)

print(f"\n   TF-IDF matrix (train) : {X_train_tfidf.shape}")
print(f"   TF-IDF matrix (test)  : {X_test_tfidf.shape}")
print(f"   Sample features       :
{tfidf.get_feature_names_out()[:15].tolist()}")

#  PHASE 4 — MODELLING  (CRISP-DM Step 4)

print("\n" + "—" * 60)
```

```
print("PHASE 4 — MODELLING")
print("—" * 60)

# 4.1  Define Base Models
#
# Model 1: Multinomial Naive Bayes
#   - Probabilistic classifier based on Bayes' theorem
#   - Works well with TF-IDF sparse matrices
#   - Fast training, good baseline for text classification
#
# Model 2: Logistic Regression
#   - Linear classifier optimised with multinomial log-loss
#   - Handles high-dimensional sparse text features well
#   - Provides probability estimates per class
#
# Model 3: Support Vector Machine (LinearSVC)
#   - Finds optimal hyperplane maximising margin between classes
#   - Excellent for high-dimensional text data
#   - Strong performance on NLP classification tasks

base_models = {
    "Naive Bayes": MultinomialNB(),
    "Logistic Regression": LogisticRegression(
        multi_class='multinomial', solver='lbfgs',
        max_iter=1000, random_state=42
    ),
    "Support Vector Machine": LinearSVC(
        max_iter=3000, random_state=42
    )
}

#  4.2  Hyperparameter Tuning — GridSearchCV
# GridSearchCV exhaustively searches over a parameter grid
# using Stratified K-Fold cross-validation (k=3) to prevent
```

```
# data leakage and ensure fair evaluation on unseen folds.

print(" Hyperparameter Tuning with GridSearchCV (3-Fold CV)...")
print("   (Expected time: 5-10 minutes)\n")

param_grids = {
    "Naive Bayes": {
        'alpha': [0.01, 0.1, 0.5, 1.0]
    },
    "Logistic Regression": {
        'C'       : [0.1, 1.0, 5.0, 10.0],
        'max_iter': [1000]
    },
    "Support Vector Machine": {
        'C': [0.1, 0.5, 1.0, 5.0]
    }
}

cv_strategy  = StratifiedKFold(n_splits=3, shuffle=True, random_state=42)
tuned_models = {}

for name, model in base_models.items():
    print(f"   Tuning: {name}")
    t0 = time.time()

    grid_search = GridSearchCV(
        estimator=model,
        param_grid=param_grids[name],
        scoring='f1_weighted',
        cv=cv_strategy,
        n_jobs=-1,
        verbose=0
    )
    grid_search.fit(X_train_tfidf, y_train)
```

```
    best_model  = grid_search.best_estimator_
    best_params = grid_search.best_params_
    best_score  = grid_search.best_score_
    elapsed     = time.time() - t0

    tuned_models[name] = {
        'model'      : best_model,
        'best_params': best_params,
        'cv_score'   : best_score
    }

    print(f"     Best params : {best_params}")
    print(f"     CV F1       : {best_score:.4f}  |  Time:
{elapsed:.1f}s\n")

print(" Hyperparameter tuning complete!")

# 4.3  Train Tuned Models & Predict
results = {}

print("\n Training final tuned models on full training set...\n")

for name, info in tuned_models.items():
    model  = info['model']
    y_pred = model.predict(X_test_tfidf)

    acc     = accuracy_score(y_test, y_pred)
    f1_mac  = f1_score(y_test, y_pred, average='macro')
    f1_wt   = f1_score(y_test, y_pred, average='weighted')
    prec    = precision_score(y_test, y_pred, average='weighted')
    rec     = recall_score(y_test, y_pred, average='weighted')
    report  = classification_report(
                  y_test, y_pred,
                  target_names=['Negative', 'Neutral', 'Positive']
```

```
            )
    cm      = confusion_matrix(y_test, y_pred)

    results[name] = {
        'model'      : model,
        'best_params': info['best_params'],
        'cv_score'   : info['cv_score'],
        'y_pred'     : y_pred,
        'accuracy'   : acc,
        'f1_macro'   : f1_mac,
        'f1_weighted': f1_wt,
        'precision'  : prec,
        'recall'     : rec,
        'report'     : report,
        'cm'         : cm
    }

    print(f"   {name}")
    print(f"     Accuracy     : {acc:.4f}")
    print(f"     F1-Macro     : {f1_mac:.4f}")
    print(f"     F1-Weighted  : {f1_wt:.4f}")
    print(f"     Precision    : {prec:.4f}")
    print(f"     Recall       : {rec:.4f}\n")

#  PHASE 5 — EVALUATION  (CRISP-DM Step 5)
print("—" * 60)
print("PHASE 5 — EVALUATION")
print("—" * 60)

# 5.1  Summary Comparison Table
print("\n" + "=" * 72)
print("                    MODEL COMPARISON SUMMARY")
print("=" * 72)
print(f"{'Model':<28} {'Accuracy':>9} {'F1-Mac':>8} {'F1-Wt':>8} "
```

```
      f"{'Prec':>8} {'Recall':>8}")
print("-" * 72)
for name, r in results.items():
    print(f"{name:<28} {r['accuracy']:>9.4f} {r['f1_macro']:>8.4f} "
          f"{r['f1_weighted']:>8.4f} {r['precision']:>8.4f}
{r['recall']:>8.4f}")
print("=" * 72)

# Best model
best_name = max(results, key=lambda n: results[n]['f1_weighted'])
print(f"\n Best Model: {best_name}  "
      f"(F1-Weighted = {results[best_name]['f1_weighted']:.4f})")

#  5.2  Detailed Classification Report
print(f"\nDetailed Classification Report — {best_name}:")
print(results[best_name]['report'])

#  5.3  Confusion Matrices
fig, axes = plt.subplots(1, 3, figsize=(18, 5))
CLASS_LABELS = ['Negative', 'Neutral', 'Positive']

for ax, (name, r) in zip(axes, results.items()):
    cm_pct = (r['cm'].astype('float') /
              r['cm'].sum(axis=1)[:, np.newaxis] * 100)
    sns.heatmap(cm_pct, annot=True, fmt='.1f', cmap='Blues',
                xticklabels=CLASS_LABELS,
                yticklabels=CLASS_LABELS, ax=ax,
                linewidths=0.5, linecolor='white')
    ax.set_title(f'{name}\nAcc: {r["accuracy"]:.4f} | F1:
{r["f1_weighted"]:.4f}',
                 fontsize=11, fontweight='bold')
    ax.set_xlabel('Predicted Label', fontsize=10)
    ax.set_ylabel('True Label', fontsize=10)

plt.suptitle('Confusion Matrices — % of True Class',
```

```
             fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('fig5_confusion_matrices.png', dpi=150, bbox_inches='tight')
plt.show()
print("    Fig 5 saved: fig5_confusion_matrices.png")

# 5.4  Model Comparison Bar Chart
model_names = list(results.keys())
metrics     = {
    'Accuracy'    : [results[n]['accuracy']    for n in model_names],
    'F1-Macro'    : [results[n]['f1_macro']    for n in model_names],
    'F1-Weighted' : [results[n]['f1_weighted'] for n in model_names],
    'Precision'   : [results[n]['precision']   for n in model_names],
    'Recall'      : [results[n]['recall']      for n in model_names],
}

x      = np.arange(len(model_names))
width  = 0.15
colors = ['#3498db', '#e74c3c', '#2ecc71', '#9b59b6', '#f39c12']

fig, ax = plt.subplots(figsize=(13, 6))
for i, (metric, vals) in enumerate(metrics.items()):
    offset = (i - len(metrics)/2) * width + width/2
    bars   = ax.bar(x + offset, vals, width,
                    label=metric, color=colors[i], alpha=0.88)
    for bar in bars:
        ax.annotate(f'{bar.get_height():.3f}',
                    xy=(bar.get_x() + bar.get_width()/2,
                        bar.get_height()),
                    xytext=(0, 3), textcoords='offset points',
                    ha='center', va='bottom', fontsize=7.5)

ax.set_xlabel('Model', fontsize=12)
ax.set_ylabel('Score', fontsize=12)
```

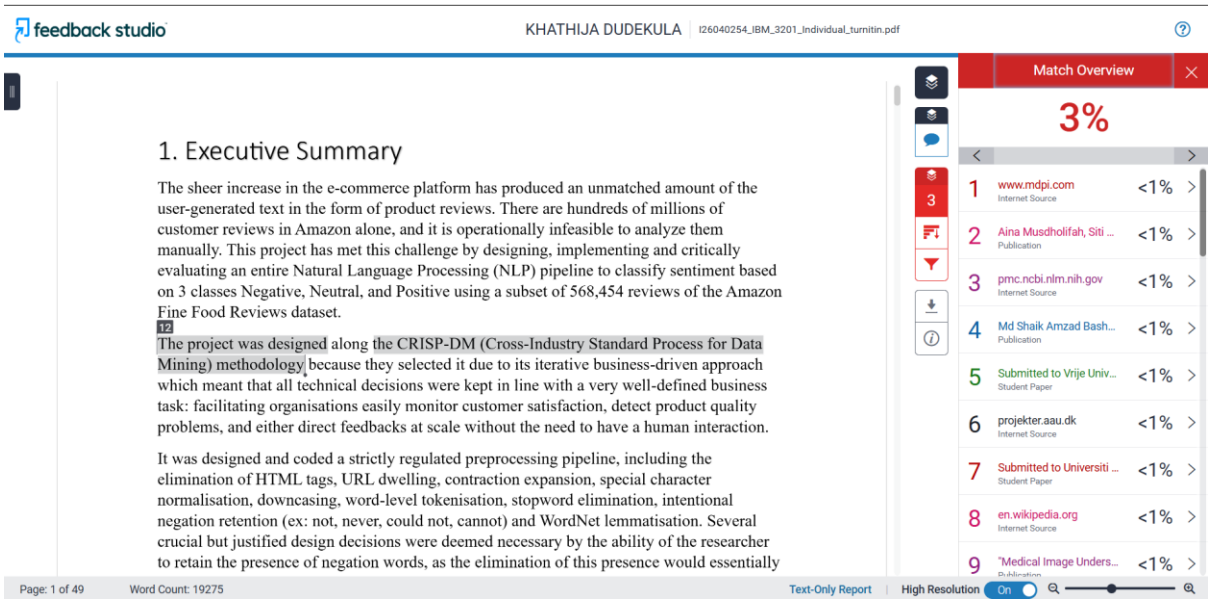
```
ax.set_title('Model Comparison Across All Evaluation Metrics',
             fontsize=13, fontweight='bold')
ax.set_xticks(x)
ax.set_xticklabels(['Naive Bayes', 'Logistic Reg.', 'SVM'], fontsize=11)
ax.set_ylim(0.3, 1.10)
ax.legend(fontsize=10, loc='lower right')
ax.axhline(y=0.80, color='grey', linestyle='--',
           alpha=0.5, label='Target (0.80)')
plt.tight_layout()
plt.savefig('fig6_model_comparison.png', dpi=150, bbox_inches='tight')
plt.show()
print("    Fig 6 saved: fig6_model_comparison.png")

# 5.5  Per-Class F1-Score Heatmap
heatmap_data = []

for name in model_names:
    row    = {}
    lines  = results[name]['report'].split('\n')
    for line in lines[2:5]:
        parts = line.split()
        if len(parts) >= 4:
            row[parts[0]] = float(parts[3])
    heatmap_data.append([
        row.get('Negative', 0),
        row.get('Neutral',  0),
        row.get('Positive', 0)
    ])

hm_df = pd.DataFrame(
    heatmap_data,
    index=['Naive Bayes', 'Logistic Reg.', 'SVM'],
    columns=['Negative', 'Neutral', 'Positive']
)
```

```
plt.figure(figsize=(8, 3.5))
sns.heatmap(hm_df, annot=True, fmt='.3f', cmap='RdYlGn',
            vmin=0.4, vmax=1.0, linewidths=0.5, linecolor='white')
plt.title('Per-Class F1-Score Heatmap', fontsize=13, fontweight='bold')
plt.xlabel('Sentiment Class')
plt.ylabel('Model')
plt.tight_layout()
plt.savefig('fig7_f1_heatmap.png', dpi=150, bbox_inches='tight')
plt.show()
print("    Fig 7 saved: fig7_f1_heatmap.png")

# 5.6  Hyperparameter Tuning Results Chart
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

for ax, (name, r) in zip(axes, results.items()):
    param_name  = list(param_grids[name].keys())[0]
    param_vals  = param_grids[name][param_name]
    mean_scores = []

    for val in param_vals:
        params = {param_name: val}
        if name == "Logistic Regression":
            params['max_iter'] = 1000
        temp_model = base_models[name].__class__(**params)
        scores = cross_val_score(
            temp_model, X_train_tfidf, y_train,
            cv=3, scoring='f1_weighted', n_jobs=-1
        )
        mean_scores.append(scores.mean())

    ax.plot([str(v) for v in param_vals], mean_scores,
            marker='o', color=list(CLASS_COLORS.values())[
                list(results.keys()).index(name)],
```

```
            linewidth=2, markersize=8)
    ax.set_title(f'{name}\nBest {param_name}='
                 f'{r["best_params"][param_name]}', fontsize=11)
    ax.set_xlabel(param_name)
    ax.set_ylabel('F1-Weighted (CV)')
    ax.set_ylim(0.5, 1.0)
    ax.grid(True, alpha=0.3)

plt.suptitle('GridSearchCV — Hyperparameter Tuning Results',
             fontsize=13, fontweight='bold')
plt.tight_layout()
plt.savefig('fig8_hyperparameter_tuning.png', dpi=150, bbox_inches='tight')
plt.show()
print("    Fig 8 saved: fig8_hyperparameter_tuning.png")

#  PHASE 6 — DEPLOYMENT  (CRISP-DM Step 6)

print("\n" + "—" * 60)
print("PHASE 6 — DEPLOYMENT")
print("—" * 60)

def predict_sentiment(text, model_name=None):
    """
    Predict 3-class sentiment for any input text.

    Parameters : text       (str) — raw review text
                 model_name (str) — model to use (default: best model)
    Returns    : dict with label, class name, and confidence note
    """
    if model_name is None:
        model_name = best_name

    cleaned  = preprocess_text(text)
    vec      = tfidf.transform([cleaned])
```

```
    pred     = results[model_name]['model'].predict(vec)[0]


    emoji_map = {0: " Negative", 1: " Neutral", 2: " Positive"}
    return {
        'sentiment'   : emoji_map[pred],
        'label'       : int(pred),
        'model_used'  : model_name,
        'cleaned_text': cleaned
    }


# Demo predictions
test_reviews = [
    "This is absolutely the best granola I have ever tasted. Will buy
again!",
    "Product arrived on time but the taste is just okay. Nothing special.",
    "Complete waste of money. Tastes awful and packaging was damaged.",
    "Not bad for the price. I expected better flavour but it does the
job.",
    "My dog loves these treats! She goes crazy every time I open the bag.",
    "Disgusting smell, terrible aftertaste. Do not buy this product.",
]


print(f"\n Sentiment Predictions using best model: {best_name}\n")
print("—" * 60)


for review in test_reviews:
    result = predict_sentiment(review)
    print(f"Review   : {review[:75]}...")
    print(f"Verdict  : {result['sentiment']}")
    print()


#  Final Summary
print("=" * 60)
print("  FINAL SUMMARY")
print("=" * 60)
```

```
print(f"  Dataset    : Amazon Fine Food Reviews")
print(f"  Task       : 3-Class Sentiment Analysis")
print(f"  Total docs : {len(cleaned_texts):,}")
print(f"  Features   : TF-IDF (unigrams + bigrams, 15k features)")
print()
for name, r in results.items():
    marker = " ← BEST" if name == best_name else ""
    print(f"  {name:<28} F1={r['f1_weighted']:.4f}{marker}")

print()
print("  Charts saved (use in report):")
charts = [
    "fig1_class_distribution.png",
    "fig2_text_length.png",
    "fig3_wordclouds.png",
    "fig4_top_words.png",
    "fig5_confusion_matrices.png",
    "fig6_model_comparison.png",
    "fig7_f1_heatmap.png",
    "fig8_hyperparameter_tuning.png",
]
for c in charts:
    print(f"     {c}")

print()
print(" Pipeline complete! You are ready to write your report.")
print("=" * 60)
```



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1. Executive Summary

The sheer increase in the e-commerce platform has produced an unmatched amount of the user-generated text in the form of product reviews. There are hundreds of millions of customer reviews in Amazon alone, and it is operationally infeasible to analyze them manually. This project has met this challenge by designing, implementing and critically evaluating an entire Natural Language Processing (NLP) pipeline to classify sentiment based on 3 classes Negative, Neutral, and Positive using a subset of 568,454 reviews of the Amazon Fine Food Reviews dataset.

The project was designed along the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology because they selected it due to its iterative business-driven approach which meant that all technical decisions were kept in line with a very well-defined business task: facilitating organisations easily monitor customer satisfaction, detect product quality problems, and either direct feedbacks at scale without the need to have a human interaction.

It was designed and coded a strictly regulated preprocessing pipeline, including the elimination of HTML tags, URL dwelling, contraction expansion, special character normalisation, downcasing, word-level tokenisation, stopword elimination, intentional negation retention (ex: not, never, could not, cannot) and WordNet lemmatisation. Several crucial but justified design decisions were deemed necessary by the ability of the researcher to retain the presence of negation words, as the elimination of this presence would essentially

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